
THE SHAPE OF ATTRACTION IN UMAP: EXPLORING THE EMBEDDING FORCES IN DIMENSIONALITY REDUCTION

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ABSTRACT

Uniform manifold approximation and projection (UMAP) is among the most widely used members of the broad family of neighbor embedding methods. Like its relatives, it relies on attractive and repulsive forces among high-dimensional data points to obtain a low-dimensional embedding, a principle that spans essentially the entire class of modern dimensionality reduction algorithms. In this paper, we analyze these forces to reveal how they govern cluster formation and visualization across such methods, using UMAP as a representative case and comparing it to a range of contemporaries. We find that repulsion emphasizes differences, controlling cluster boundaries and inter-cluster distance, whereas attraction plays a more subtle role: the attractive tension between points can manifest simultaneously as attraction and repulsion in the lower-dimensional mapping. This dual behavior explains the general need for learning rate annealing and motivates the distinct treatments that these algorithms give to attractive and repulsive terms. Moreover, by modifying attraction, we improve the consistency of cluster formation under random initialization, a benefit that transfers across datasets and methods. Overall, our analysis offers a unifying perspective that makes UMAP and the wider family of embedding methods more interpretable, more robust, and more accurate.

1 Introduction

Across virtually every scientific and industrial domain, the current era is characterized by a deluge of high-dimensional data. Dimensionality reduction (DR) techniques have emerged as general-purpose tools for exploratory analysis of such data, enabling practitioners to visualize the underlying structure regardless of where the data originate. Among the most widely adopted of these methods, t -distributed stochastic neighbor embedding (Maaten and Hinton, 2008) and uniform manifold approximation and projection (UMAP) (McInnes et al., 2018) are grounded in the attraction-repulsion dynamics that bring similar data points closer while pushing dissimilar ones apart. As unsupervised algorithms, these methods do not rely on labeled data; instead, they identify and preserve the intrinsic structure of high-dimensional data by leveraging local (attractive) and global (repulsive) relationships (forces). This label-free flexibility makes them broadly applicable across an expanding array of tasks, including clustering (Becht et al., 2019), exploratory data analysis (Fleischer and Islam, 2020), anomaly detection in settings such as semiconductor manufacturing (Fan et al., 2021), visual search (González-Márquez et al., 2024), time series analysis (Altin and Cakir, 2024), studying representation convergence (Huh et al., 2024), and outlier image detection (Islam and Fleischer, 2024), and more generally any setting where visualizing hidden patterns in unlabeled data is critical and meaningful. By learning the embeddings in a data-driven, label-free manner, DR exemplifies the power of unsupervised methods to distill complex data of many kinds into easily interpretable forms.

Building upon the same attraction-repulsion principle, a steady stream of newer methods has emerged (Amid and Warmuth, 2019; Agrawal et al., 2021; Wang et al., 2021; Narayan et al., 2021; Yang et al., 2022; Wang et al., 2025; Kury et al., 2025), each designed to emphasize specific aspects of the data while sharing this common foundation. Despite their widespread relevance across applications, these methods often rely on heuristic practices that may fail to give meaningful interpretations. Moreover, DR inevitably introduces distortions (Chari and Pachter, 2023). It is therefore imperative to develop a deeper, method-agnostic understanding of the algorithms so that practitioners can better interpret the embeddings, avoid spurious structures, and optimize performance across the many domains in which these methods are deployed. In practice, such algorithms achieve compact clusters using a variety of techniques, including specific initialization, learning rate scheduling, and kernel function tuning. However, the underlying dynamics of the attractive and repulsive forces that drive cluster formation have not been thoroughly investigated in a general way.

Furthermore, the essential tunable parameters are concealed within abstract functional forms, making the algorithms harder to explain.

In this paper, we decompose the forces into their constituent parts and extract the functional shapes of attraction and repulsion, a decomposition that applies to any method built on attraction-repulsion dynamics. We find that the necessity of learning rate annealing, the challenge of providing consistent output under random initialization, and the origin of cluster formation all rely on attraction. Repulsive forces, by contrast, primarily govern inter-cluster distances. Our specific contributions are:

1. We formulate attraction and repulsion shapes from the attractive and repulsive forces, establish general conditions for contraction and expansion of distance, and provide a fresh, broadly applicable perspective on this class of algorithms (Section 4).
2. We show that the attraction shape of UMAP gives rise to the counterintuitive phenomenon of both contraction and expansion of distance. By comparing attraction shapes across different algorithms, we discuss how attraction influences the learning rate annealing scheme in general (Section 5).
3. We modify the attraction shape to improve the consistency of embedding under random initialization, and show that this improvement holds across datasets, indicating the encoding of a unique underlying structure (Section 5.1).
4. Analyzing repulsion shapes, we provide a deeper understanding of how cluster formation and inter-cluster distance are regulated (Section 5.2).
5. We compare attraction and repulsion shapes of UMAP, Parametric UMAP, and NEG- t -SNE, unveil the similarities and distinctions among them, and characterize the stability of these algorithms as instances of a broader pattern (Section 6).

We center the main text on UMAP, now a de facto standard across many fields, and bring in other algorithms where relevant to underscore the generality of our findings. For further details on the range of algorithms covered, see the appendix. The codes used in the research will be released upon acceptance.

2 Related Works

The broad lineage of modern iterative graph-based neighbor embedding algorithms can be traced to stochastic neighbor embedding (SNE) (Hinton and Roweis, 2002) and its extension using the t -distribution (t -SNE) (Maaten and Hinton, 2008). Both methods use a dense graph in which each point in a dataset has a pairwise relation with all the others, regardless of whether they are similar to each other or not. Moreover, the weights of the graphs are normalized to give a notion of probability distribution. Other concurrent methods, including locally linear embedding (Roweis and Saul, 2000) and Laplacian Eigenmaps (LE) (Belkin and Niyogi, 2002), instead used a k -nearest neighbor (k -NN) graph of pairwise interactions. Known as spectral methods, these algorithms rely on eigenvalue decomposition. Subsequent work by Tang et al. (Tang et al., 2016) incorporated the k -NN graph in an iterative approach and removed normalization in the lower dimension. This direction was carried further by McInnes et al. (2018) in UMAP, where the normalization step was removed altogether (in both high and low dimensions) and the embedding was obtained using pairwise interactions alone. The optimization steps use an explicit attractive force to preserve the local neighborhood and a repulsive force to keep dissimilar points apart. Building on these shared foundations, a growing family of methods such as PaCMAP (Wang et al., 2021) and NEG- t -SNE (Damrich et al., 2023) has since been proposed. For a recent survey spanning this landscape, see de Bodt et al. (2025).

There has been considerable progress in understanding and explaining the relationships among these algorithms, and much of it points toward common underlying principles. An early analysis of SNE found that if the data are well-clustered in the original space, then they remain well-clustered in the embedding space (Shaham and Steinerberger, 2017). A similar analysis for t -SNE by Linderman and Steinerberger (2019) showed that the number of clusters in the embedding space is a lower bound on the number of clusters in the original space. This line was extended in further characterizations (Arora et al., 2018; Cai and Ma, 2022; Linderman and Steinerberger, 2022). Since t -SNE and UMAP originate from the same underlying framework (but produce drastically different visualizations), a major undertaking in the literature has been to uncover the connection between them, and more broadly among related methods (Böhm et al., 2022; Damrich et al., 2023; Draganov et al., 2023). Böhm et al. (2022) theorized that methods such as Laplacian Eigenmaps, ForceAtlas2 (Jacomy et al., 2014), UMAP, and t -SNE are all samples from the same underlying spectrum. Indeed, LE and t -SNE are connected through the early exaggeration phase (Cai and Ma, 2022), and UMAP and t -SNE can be related through contrastive estimation (Damrich et al., 2023). Around the same time, Hu et al. (2023) independently discovered the relation between contrastive learning and SNE. More recent approaches offer

a probabilistic perspective (Ravuri et al., 2023; Ravuri and Lawrence, 2024), employ kernel techniques (Draganov and Dohn, 2023), and utilize information geometry (Kolpakov and Rocke, 2024) to explain dimensionality reduction, reflecting a broad and ongoing effort to place these methods within a common conceptual frame.

3 Uniform Manifold Approximation and Projection

UMAP constructs a high-dimensional graph of the dataset $X = \{x_i \in \mathbb{R}^n | i = 1, \dots, N\}$ by using a pairwise relation: $p_{i,j} = f_h(d(x_i, x_j))$, typically, $\in [0, 1]$ (for details, see Appendix H).

The graph of the low-dimensional representation $Y = \{y_i \in \mathbb{R}^d | i = 1, \dots, N\}$ is given by a differentiable function

$$q_{ij} = \frac{1}{1 + a(\|y_i - y_j\|_2^2)^b}, \quad (1)$$

where the parameters a and b determine the density of the mapping and are chosen by fitting q_{ij} to

$$\Psi(d_{ij}) = \begin{cases} 1 & \text{if } \|y_i - y_j\|_2 < m_d, \\ \exp(-(d_{ij} - m_d)) & \text{otherwise} \end{cases}, \quad (2)$$

where m_d regulates the distance between the two nearest low-dimensional points.

UMAP aims to minimize the following cross-entropy loss function:

$$\mathcal{L} = \sum_{i,j} (-p_{ij} \log(q_{ij}) - (1 - p_{ij}) \log(1 - q_{ij})). \quad (3)$$

The first term provides an attractive force and the second term provides a repulsive force, a split that is characteristic of neighbor embedding methods more broadly. Instead of optimizing every point in each iteration, UMAP takes the negative sampling approach (Mikolov et al., 2013; Tang et al., 2016). For each edge with $p_{ij} > 0$, called a positive edge, several edges are sampled at random, called negative edges. The attractive force is applied on the positive edge:

$$y_i^{t+1} = y_i^t + \lambda \nabla_{y_i^t} \log(q_{ij}), \quad (4)$$

$$y_j^{t+1} = y_j^t + \lambda \nabla_{y_j^t} \log(q_{ij}), \quad (5)$$

and the repulsive force is applied on the negative edges:

$$y_i^{t+1} = y_i^t + \lambda \nabla_{y_i^t} \log(1 - q_{ij}), \quad (6)$$

where $\lambda(> 0)$ is the learning rate and t is the step number. Note that y_j is not updated for negative edges. For a detailed analysis of UMAP's loss, see Damrich and Hamprecht (2021).

4 Attraction and Repulsion Shapes

The action of the updates (4, 6) can be simplified by decomposing the gradients $\nabla_{y_i^t} \log(q_{ij})$ ($\nabla_{y_i^t} \log(1 - q_{ij})$) into a scalar coefficient that depends on the distance $\|y_i - y_j\|_2$ and acts on the vector $(y_i - y_j)$. We call this scalar coefficient the attraction (repulsion) shape. Although we use UMAP as a concrete running example, the formalism is general: it applies to any method that relies on attraction and repulsion, and we exploit this generality throughout the paper.

By writing $\nabla_{y_i^t} \log(q_{ij}) = f_a(\zeta^t)(y_i^t - y_j^t)$, where $\zeta = \|y_i - y_j\|_2$, we can update the equations of a positive edge as

$$y_i^{t+1} = y_i^t + \lambda f_a(\zeta^t)(y_i^t - y_j^t), \quad (7)$$

$$y_j^{t+1} = y_j^t - \lambda f_a(\zeta^t)(y_i^t - y_j^t). \quad (8)$$

Here, $f_a : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\leq 0}$ is the attraction shape, and we use the fact that for Euclidean metric, $\nabla_{y_i^t} \log(q_{ij}) = -\nabla_{y_j^t} \log(q_{ij})$. Similarly, we reformulate the update equation of a negative edge as

$$y_i^{t+1} = y_i^t + \lambda f_r(\zeta^t)(y_i^t - y_j^t), \quad (9)$$

$$y_j^{t+1} = y_j^t, \quad (10)$$

where $f_r : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is the repulsion shape.

Such decompositions have appeared previously, e.g., in (McInnes et al. 2018, Agrawal et al. 2021, Draganov and Dohn 2023), though their formulations and uses vary widely. The original UMAP paper (McInnes et al. 2018) used it as a computational trick for fast processing, while (Draganov and Dohn 2023) used it to compare different algorithms from a kernel perspective. In both cases, the primary focus was computing the derivative. Here, in contrast, we treat the decompositions as independent functions that can take a variety of forms, which lets us reason about attraction and repulsion in a method-independent way. A related approach was used earlier by (Agrawal et al. 2021), but they expressed the decomposition in terms of a scalar coefficient and a unit vector $((y_i - y_j)/\|y_i - y_j\|_2)$ to emphasize the magnitude of the forces. The discussion below shows that our shape decomposition is more illuminating than magnitude alone, and that it generalizes readily across methods.

4.1 Conditions for Attraction and Repulsion

In this section, we establish general conditions for attraction and repulsion directly from the update equations (7)-(10); because these conditions are stated in terms of the shapes alone, they hold for any method sharing this update structure. The following proposition characterizes the contraction of distance between the pair y_i^t and y_j^t of a positive edge:

Proposition 4.1. *The update Eqs. (7) and (8) provide a contraction of distance ($\|y_i^{t+1} - y_j^{t+1}\| < \|y_i^t - y_j^t\|$) if $-1 < \lambda f_a < 0$.*

Here, λf_a works as the effective attraction shape. In most cases, however, f_a alone is enough to draw meaningful conclusions about the embeddings. For a negative edge, the following proposition characterizes the expansion of the distance between the pair y_i^t and y_j^t :

Proposition 4.2. *The update Eqs. (9) and (10) provide an expansion of distance ($\|y_i^{t+1} - y_j^{t+1}\| > \|y_i^t - y_j^t\|$) if $f_r > 0$.*

Note that including a symmetric term $-\lambda f_r(\zeta)(y_i^t - y_j^t)$ in Eq. (10) does not alter the conclusion of this proposition. More generally, these conditions provide a per-iteration certificate for guaranteed contraction/expansion that applies to any shape, and they should not be conflated with the learning rate tuning/decay mechanism: one can design shapes for which contraction and expansion do not rely on learning rate decay at all.

Figure 1(a) shows the effect of different values of $f_a < 0$ and $f_r > 0$ on two points. The repulsion shape behaves straightforwardly, as any positive value increases the distance. The attraction shape is far more subtle, since f_a encodes both attractive and repulsive dynamics. For $f_a \in (-1.0, 0)$, the distance decreases, with a sign flip at the value $f_a = -0.5$ of maximum attraction (coincident points). Any value below -1.0 causes the distance to increase. Although these forces act locally on individual pairs, they collectively shape the global structure of the embedding, and this local-to-global interplay recurs across the methods we study.

5 Analysis of UMAP in terms of Attraction and Repulsion Shape

Using the gradient decomposition and the distance form (1), the attraction and repulsion shapes are given by

$$f_a^U(\zeta) = -\frac{2ab\zeta^{2(b-1)}}{1 + a\zeta^{2b}} \quad (11)$$

and

$$f_r^U(\zeta) = \frac{2b}{\zeta^{2b}(1 + a\zeta^{2b})}, \quad (12)$$

respectively. This section focuses on the default shapes of UMAP, which are controlled primarily by the parameters a and b , and draws out the insights gained by perturbing them; as we show later, the resulting conclusions extend naturally to other methods. The discussion below holds generally for $b \leq 1$, where the attraction shape is strictly increasing and thus invertible (for the derivation, see Appendix A).

Figure 1(b) shows the default attraction shape of UMAP ($a = 1.58, b = 0.89$). It becomes unbounded (approaches ∞) as $\zeta \rightarrow 0$. As predicted by Proposition 4.1 the transition from contraction to expansion occurs when λf_a^U crosses -1 as ζ approaches 0. Since the attraction shape is invertible, we can identify the distance at this transition, $\zeta_{-1} = (\lambda f_a^U)^{-1}(-1)$, as the minimum distance for contraction due to attractive updates. In effect, $\zeta > \zeta_{-1}$ causes contraction while $0 < \zeta < \zeta_{-1}$ causes expansion, contradicting the common intuition that attractive updates consistently bring points closer together. This counterintuitive behavior is a general consequence of the shape, not a quirk of any particular dataset.

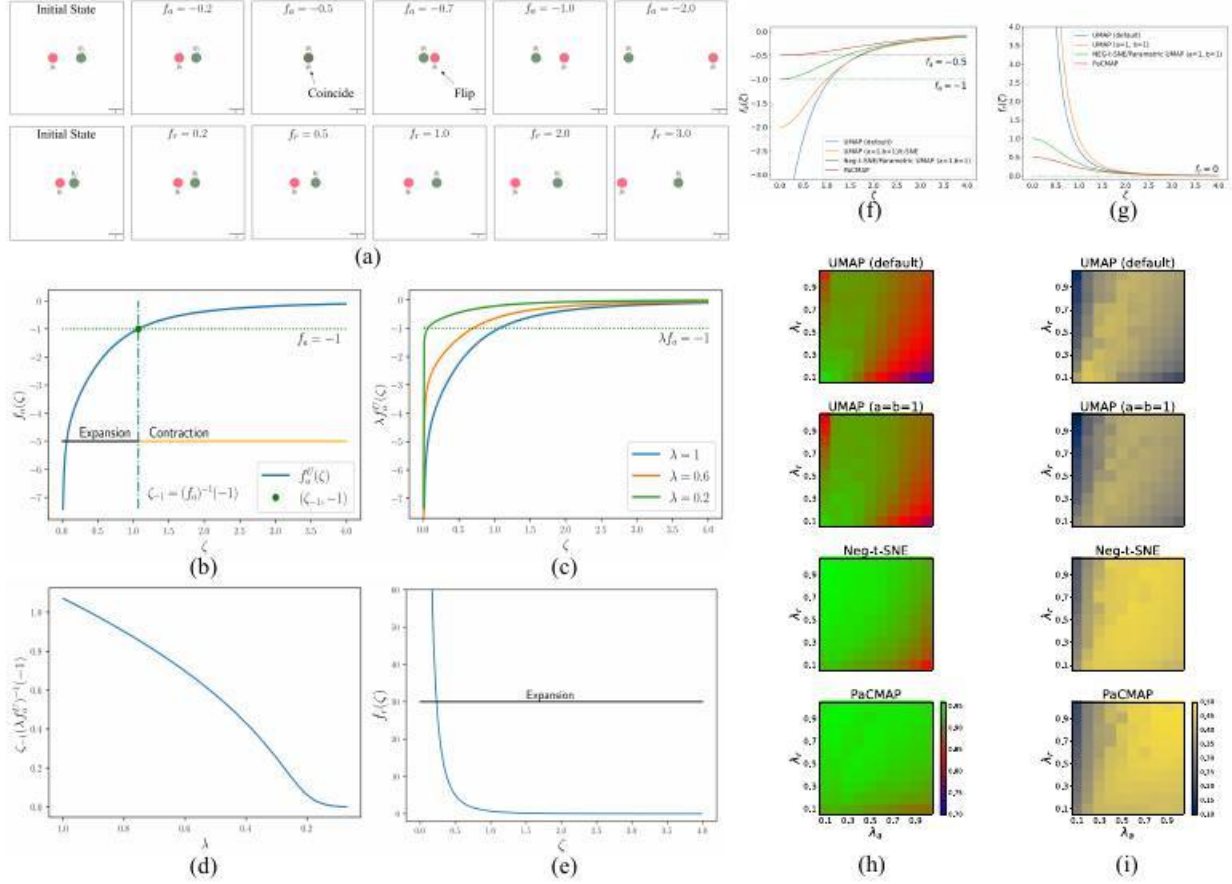


Figure 1: Attraction and repulsion shapes, illustrated on UMAP and compared across methods. (a) Effect of different values of f_a (top) and f_r (bottom) on a pair. (b) Attraction shape of UMAP. (c) Effective attraction shape (λf_a) for various learning rates λ . (d) Minimum distance for contraction (ζ_{-1}) as λ decreases. (e) Repulsion shape of UMAP. (f) Attraction and (g) repulsion shapes across a range of embedding methods. (h) Trustworthiness and (i) Silhouette Score for these methods as the constant learning rate is varied for attraction (λ_a) and repulsion (λ_r) independently. Default UMAP parameters: $a = 1.58$ and $b = 0.89$.

When ζ_{-1} is high, neighboring points oscillate between contraction and expansion, and the clusters appear fuzzy. For the sharpest boundaries, then, the goal of optimization can be recast as achieving the limit $\zeta_{-1} \rightarrow 0$. As a result, UMAP’s learning rate schedule requires annealing to zero (Figs. 1(c,d)); as we show below, whether a method needs such annealing follows directly from where its attraction shape sits relative to this threshold. The repulsion shape (Eq. 12), by contrast, is always positive and satisfies Proposition 4.2 f_r^U approaches 0 as $\zeta \rightarrow \infty$ and approaches ∞ as $\zeta \rightarrow 0$ (Fig. 1(e)).

Comparing attraction shapes across algorithms reveals a clear pattern: only UMAP and t -SNE must contend with $f_a < -1$ (and consequently $\zeta_1 > 0$ at $\lambda = 1$, Fig. 1(f)). t -SNE addresses this by weighting the updates with the corresponding p_{ij} values, whereas UMAP relies on learning rate annealing. Methods such as NEG- t -SNE and PaCMAP instead keep f_a naturally within $[-1, 0]$ (thereby satisfying Proposition 4.1 for $\lambda \in [0, 1]$ with $\zeta_{-1} = 0$). PaCMAP’s weighted f_a (< 0.5 always) goes further and prevents any flips during attraction. Turning to repulsion, the same cross-method comparison shows that only UMAP has to handle large values at small distances (Fig. 1(g)); the shape is unbounded, though during optimization the values are often clipped). This viewpoint thus organizes a diverse set of methods along a small number of shared axes; for additional details on these and related methods, see Appendix G.

We illustrate the effect of these attraction/repulsion choices using an MNIST embedding, applying separate constant learning rates for attraction (λ_a) and repulsion (λ_r) and varying their values (Figs. 1(h,i)); the same behavior recurs on the other datasets we examine. For UMAP, a lower value of λ_a (< 0.5 , and consequently $\zeta^{-1} \simeq 0$) is preferred for better embedding (trustworthiness (T) (Venna and Kaski 2001), a measure of structure preservation, in Fig. 1(h)) and clustering (silhouette score (SIL) (Rousseeuw 1987), a measure of cluster separation, in Fig. 1(i)), whereas the whole

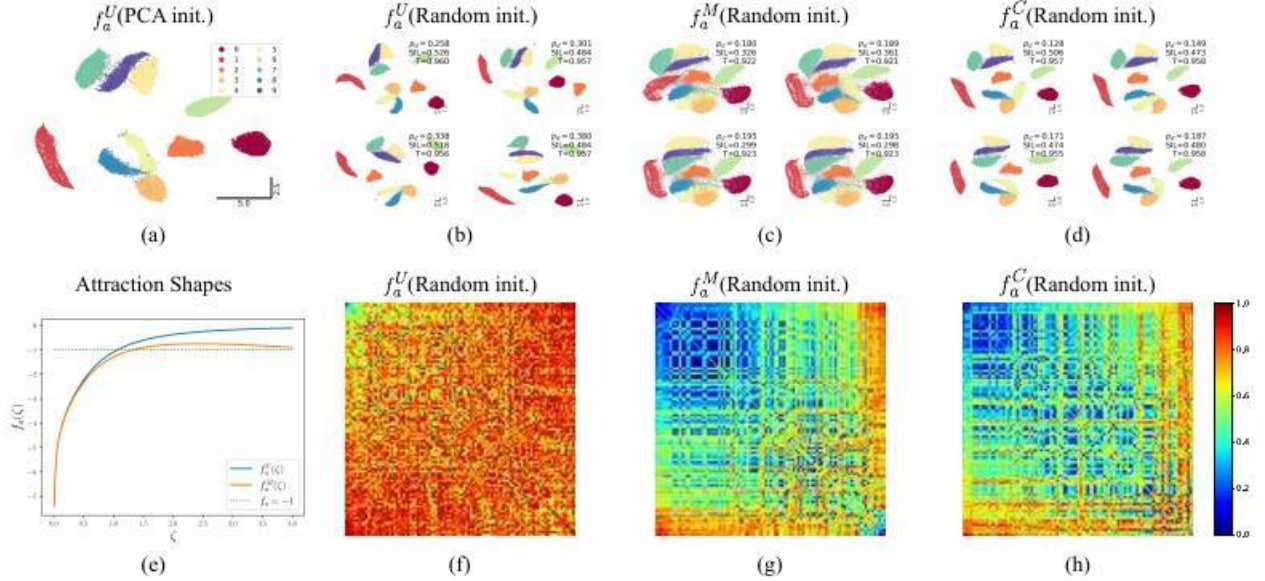


Figure 2: Effect of random initialization on different attraction shapes, shown here for the MNIST dataset as a representative case. (a) Mapping using PCA. (b-d) Four mappings with the lowest Procrustes distance (p_d) from the embedding in (a) for (b) UMAP, (c) modified, and (d) composite attraction shapes. (e) Default UMAP and modified attraction shapes. (f-h) Procrustes matrices from 100 runs of (f) UMAP (0.78 ± 0.13), (g) modified (0.49 ± 0.21), and (h) composite attraction (0.50 ± 0.20) shapes. The diagonal (i, i) entries of the Procrustes matrix are sorted by Procrustes distance (p_d) from (a), and the off-diagonal (i, j) entries correspond to p_d between the i^{th} and j^{th} mapping. The matrices and (mean $p_d \pm \text{std}$) values show that UMAP’s embeddings are not self-similar, whereas the modified and composite attraction shapes encourage scale-invariant structure — a trend that, as later sections confirm, holds across datasets.

range of λ_r ($\in [0, 1]$) is usable. For NEG- t -SNE and PaCMAP, for which $\zeta_{-1} = 0$, the whole range of parameters is effective. This confirms a general principle: UMAP relies on driving ζ_{-1} close to 0 for satisfactory embedding and clustering, which it achieves in practice through learning rate annealing, whereas $\zeta_{-1} = 0$ is the default in newer algorithms. For further discussion, results, and illustrations of these methods across different datasets, see Section 6 and Appendices B, G, and I.

5.1 Improving UMAP’s Consistency under Random Initialization

The consistency of UMAP embeddings depends on proper initialization (Kobak and Linderman 2021; Wang et al. 2021), and the same sensitivity is shared by related neighbor embedding methods. Typically, principal component analysis (PCA) of the data or spectral decomposition of the high-dimensional graph initializes the embedding, producing consistent mappings despite various sources of stochasticity. When initialized randomly, clusters often fail to form, or form in a different orientation each time the algorithm runs. If the initial distance between two points (nearest neighbors in high dimension) is large, the attractive forces become too weak to bring them closer. Known as near-sightedness (Wang et al. 2021), this phenomenon is directly visible in the attraction shape, where $|f_a^U|$ diminishes toward zero as the distance increases (i.e., $\lim_{\zeta \rightarrow \infty} |f_a^U(\zeta)| = 0$), and it is a generic feature of shapes that decay at large distances rather than a UMAP-specific artifact.

One can induce “far-sightedness” in the mapping by increasing attraction at large distances, enabling faraway neighbors to be drawn together. To test this hypothesis in a way that generalizes beyond a single shape, we modify the attraction shape of UMAP to increase the attractive force:

$$f_a^M = f_a^U - \beta\zeta, \quad (13)$$

where β is a parameter that regulates the strength of the added term (we used $\beta = 0.2$, Fig. 2(e)). This addition to the attraction shape translates into an added regularizer in the attractive term of the loss function (i.e., $\mathcal{L}^M = \mathcal{L} + \sum_{i,j} \beta/3 \|y_i - y_j\|_2^3$). Beyond attracting pairs at faraway distances, this technique enables an intermixing of points that aids convergence under random initialization (akin to early exaggeration in t -SNE). In (13), we chose the simplest linear correction; a broad family of alternatives, such as $\log \zeta$ or ζ^p ($p \in \mathbb{R}_{\geq 0}$), would serve equally well, underscoring that the effect stems from the qualitative shape rather than the specific functional form.

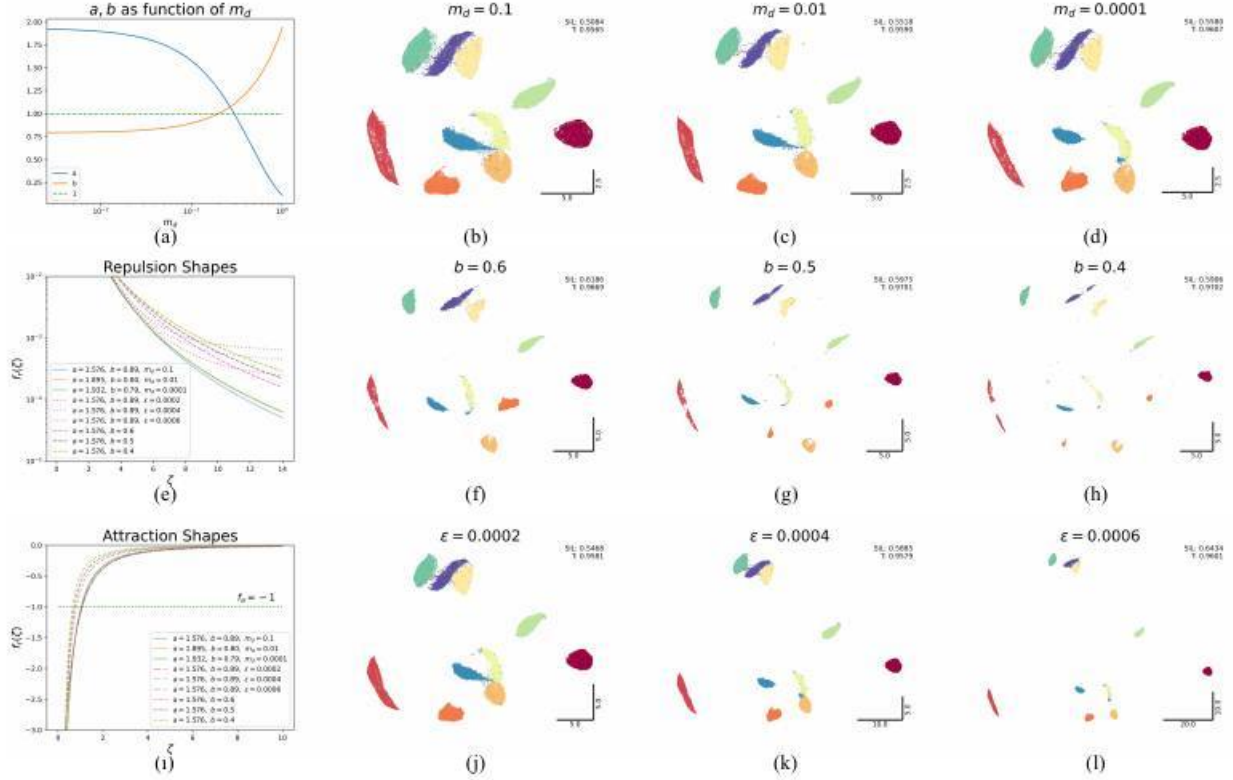


Figure 3: Control of inter-cluster distances, illustrated on the MNIST dataset. (a) Computing a, b by varying the low-dimensional distance m_d restricts exploration. (b-d) UMAP output with m_d set to 0.1, 0.01, and 0.0001, respectively, shows little improvement in cluster compactness. (e) Repulsion shapes for different parameters. (f-h) Increasing repulsion by explicitly varying b yields more compact clusters and forms new ones that were otherwise absent. (i) Attraction shapes obtained by varying parameters. (j-l) Increasing repulsion by adding a small positive value (ϵ) to the repulsion shape increases inter-cluster distance.

We also consider a composite attraction shape:

$$f_a^C = \begin{cases} f_a^M, & \text{epoch} \leq 100 \\ f_a^U, & \text{otherwise} \end{cases} \quad (14)$$

The composite shape attempts to remove any distortions introduced by f_a^M by reverting to the original UMAP. Below, we discuss the effects that these modified and composite attraction shapes have on DR from random initialization, effects that we later confirm are not specific to any one dataset.

We first created a PCA-initialized embedding of the MNIST dataset (Figure 2(a)). We then produced embeddings using random (Gaussian) initialization for each shape, repeating the experiment 100 times. To quantify the results, we use Procrustes analysis (Gower, 1975), which aligns two point clouds under scaling, translation, rotation, and reflection (for details, see Appendix D.1). Here, we align the randomly initialized embeddings to the PCA-initialized one and characterize their separation using the Procrustes distance (p_d). Figure 2(b) shows four embeddings with the lowest p_d . While the cluster shapes are consistent, their placements are not. Outputs from the modified and composite attraction shapes (Figs. 2(c) and (d), respectively) show markedly improved consistency of cluster placements.

To quantify the placements further, we consider the Procrustes matrix: the diagonal of the matrix is sorted by p_d from the PCA-initialized mapping, and the off-diagonal values are p_d between two randomly initialized mappings (for details, see Appendix D.1). This quantification is analogous to the similarity matrix (Foote, 1999). The embeddings produced by the default UMAP attraction shape are not similar to one another (Fig. 2(f)), whereas the modified (Fig. 2(g)) and composite (Fig. 2(h)) shapes are strongly similar. This indicates that UMAP encodes a unique underlying structure regardless of the initialization — a conclusion we expect to hold generally, since it follows from the shape rather than the data. In practice, however, attaining that structure in the low dimension can fall short because of the small attraction at longer distances.

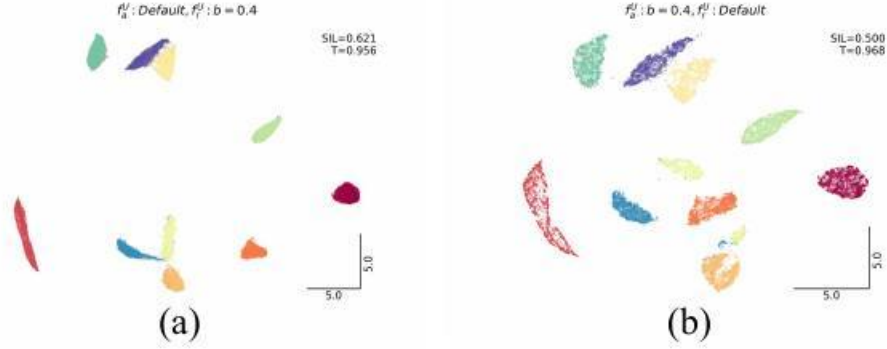


Figure 4: Embedding of the MNIST dataset with (a) the default attraction shape but a repulsion shape with $b = 0.4$ and (b) the default repulsion shape but an attraction shape with $b = 0.4$. The former reproduces the same clusters as default UMAP with increased compactness, while the latter develops new structures within clusters and forms new clusters — disentangling the distinct roles of the two forces.

5.2 Cluster Formation and Compactness

The primary controllable parameter influencing cluster formation in UMAP is the minimum distance parameter m_d (through Eq. 2). However, varying m_d restricts the exploration of different values of a and b (Fig. 3(a)). As a result, reducing m_d often yields embeddings that provide little additional benefit (Figs. 3(b-d)). The underlying reason is the limited influence that varying m_d has on the repulsion shape (Fig. 3(e)). Alternatively, one can explicitly vary the values of a and b . Decreasing a increases repulsion but decreases attraction at a faster rate (worsening near-sightedness). Decreasing b , in contrast, affords finer control (Figs. 3(e,i)). Figures 3(f-h) show the inter-cluster distance increasing and previous clusters breaking apart as b is varied to 0.6, 0.5, and 0.4, respectively. This breaking apart arises from the increasingly heavy-tailed nature of the kernel as b decreases (heavy-tailed kernels generally produce smaller and more distinct clusters (Van der Maaten and Hinton, 2008; Kobak et al., 2019; see Kobak et al., 2019 for how varying b in UMAP modulates the tail)). Although this approach separates all ten MNIST labels into distinct clusters, the relative contributions of attraction and repulsion remain hard to disentangle: changing b amplifies repulsion more than changing m_d does, but it also reshapes the attraction profile.

To quantify the effect of repulsion independently, we can keep attraction fixed and vary the other. To achieve this, we modify the repulsion shape by adding a small positive value (ε):

$$f_r^M = f_r^U + \varepsilon, \quad (15)$$

while keeping the values of a and b constant, which effectively adds a regularizer to the repulsive term of the loss function (i.e., $\mathcal{L}^M = \mathcal{L} - \sum_{i,j} \varepsilon/2 \|y_i - y_j\|_2^2$). This lets us obtain stronger repulsion than before (Fig. 3(e)). Figs. 3(j-l) show that as ε increases, the inter-cluster distances increase as well. The resulting clustering properties, however, resemble those obtained by varying m_d , and we again obtain a loose separation of all the labels as ε increases. Overall, the parameter ε leaves the attraction shape unaffected, and varying m_d effectively traces out similar attraction shapes (Fig. 3(i)), pointing to a general conclusion: cluster formation is governed predominantly by attraction.

To resolve the mystery of cluster formation, we vary either the attraction shape or the repulsion shape individually while holding the other at its default, simply by setting b to 0.4 (Fig. 4). The default attraction is unable to produce new structures or clusters in the embedding, whereas the increased repulsion ($b = 0.4$) yields smaller clusters than the original UMAP (Fig. 4(a)). Conversely, when attraction is increased by setting $b = 0.4$ with the default repulsion, the embedding develops additional structures within each cluster (Fig. 4(b)). Some of the older clusters even separate into smaller ones. This establishes a general division of labor: attraction drives cluster formation, while repulsion makes clusters more compact (depending on the repulsion shape, either by producing smaller clusters or by increasing inter-cluster distance). We show in Appendix G.3 that LocalMAP embeddings exploit this same interplay of attraction and repulsion to separate clusters, reinforcing that the principle is not confined to UMAP.

6 Comparison to NEG- t -SNE

Setting a and b to 1 is a common choice across many dimensionality reduction algorithms. This particular setting makes the gradients in a wide range of algorithms stable (or bounded) and, in turn, eases optimization. Recently, Damrich

et al. (2023) explored this for UMAP and proposed NEG- t -SNE as a solution, which, in addition to having a stable gradient, yields more compact clustering even at a constant learning rate of 1. Parametric UMAP (Sainburg et al. 2021), meanwhile, initially used the UMAP loss formulation but later adopted a numerically stable modified cross-entropy loss function (Shi et al. 2023) with a logsigmoid kernel. Analyzing both reveals that these approaches arrive at the same formulation from different starting points, a convergence that hints at a deeper commonality and leads to the following observation:

Proposition 6.1. *Neg- t -SNE is Parametric UMAP with $a = 1$ and $b = 1$.*

With $a = 1$ and $b = 1$, the attraction and repulsion shapes of UMAP are given by $f_a^U = -2/(1 + \zeta^2)$ and $f_r^U = 2/(\zeta^2(1 + \zeta^2))$, respectively. The attraction shape becomes bounded within $[-2, 0]$, with $f_a^U(0) = -2$ (Fig. 1(f)), while the repulsion shape remains essentially unchanged (i.e., unbounded as $\zeta \rightarrow 0$, Fig. 1(g)). Since $f_a^U < -1$ as $\zeta \rightarrow 0$, Proposition 4.1 and the discussion in Section 5 imply that even this unity case still requires learning rate annealing — illustrating that the annealing requirement follows from the shape rather than from any specific parameterization.

Damrich et al. (2023) compared UMAP’s negative sampling loss function from the perspective of contrastive embedding (CE) and concluded that the effective kernel of UMAP is $1/\zeta^2$. Under the CE framework, the authors introduced NEG- t -SNE by changing the kernel to $1/(1 + \zeta^2)$. Reverting to UMAP formalism, this results in the low-dimensional affinity function $q_{ij}^N = 1/(2 + \zeta^2)$. Consequently, the attraction and repulsion shapes are $f_a^N = -2/(2 + \zeta^2)$, and $f_r^N = 2/((1 + \zeta^2)(2 + \zeta^2))$, respectively (Figs. 1(f,g)). The attraction shape is bounded within $[-1, 0]$ and satisfies Proposition 4.1. Any $\lambda \in [0, 1]$ then yields contraction and avoids oscillation between expansion and contraction. Consequently, NEG- t -SNE is less sensitive to learning rate annealing, and its clusters appear less fuzzy even at a constant $\lambda = 1.0$. Moreover, the repulsion shape of NEG- t -SNE is likewise bounded within $[0, 1]$ and does not diverge as $\zeta \rightarrow 0$, so fewer points leave the clusters when sampled randomly. Whereas Damrich et al. (2023) attributed this disparity solely to the bounded repulsive forces of NEG- t -SNE versus the unbounded ones of UMAP, our shape-based analysis in Fig. 1 shows that the attraction shape is equally responsible for the stability and compactness of the clusters — a more complete account that generalizes across methods. For additional discussion of NEG- t -SNE with illustrations, see Appendix F. Together with the discussion around Fig. 4, this suggests a broadly useful capability: attraction and repulsion shapes from different methods can be freely mixed, for instance pairing the attraction shape of UMAP with the repulsion shape of NEG- t -SNE. We show several such examples by experimenting on UMAP, NEG- t -SNE, and PaCMAP in Appendix J.

7 Discussion and Conclusion

In this work, we studied how attractive and repulsive forces relate to cluster formation across neighbor embedding methods. While it is well known that attractive forces bring similar points closer together and repulsive forces push dissimilar points apart, the precise mechanisms behind these forces had not been well characterized. Here, we demystified much of the dynamics underlying cluster formation. Although the main text focuses on UMAP and NEG- t -SNE as representative cases, the results apply broadly to dimensionality reduction; we extend the same analysis to a range of other algorithms in Appendix G, confirming its generality.

We treated the attraction and repulsion coefficients as functional mappings and analyzed them through their respective shapes, a lens that transfers directly to other methods. Characterizing these shapes (Propositions 4.1 and 4.2) revealed a counterintuitive result: the attractive forces of UMAP produce repulsion (expansion of distance) at shorter distances (Fig. 1(b)); that is, instead of bringing neighboring points closer together, they push them apart. We conclude that UMAP’s learning rate schedule alleviates this phenomenon.

We also studied the initialization of the low-dimensional embedding. Similar points that start farther apart experience low attraction and never contract well enough. Our formalism offers a general way to strengthen attraction at larger distances by adding terms to the attraction shape. Across datasets, this modification produced outputs that are more consistent under random initialization. This gives us confidence that UMAP encodes a unique underlying structure, with numerical tricks merely helping to reach it faster.

Analyzing the repulsion shape revealed that higher repulsion produces larger inter-cluster distances. To exploit this, we added a small positive value to the shape, achieving stronger repulsion than is attainable by the traditional route of varying the minimum distance parameter m_d .

Finally, we examined the relation between NEG- t -SNE and UMAP. For the particular case of $a = 1$ and $b = 1$, the key difference is that NEG- t -SNE in its default form has bounded attraction ($\in [-1, 0]$) and repulsion ($\in [0, 1]$) shapes. Both propositions are therefore satisfied regardless of the learning rate, and the method is correspondingly less sensitive

to a fixed learning rate ($\in [0, 1]$). Varying a and b in NEG- t -SNE reintroduces the same numerical challenges as UMAP, showing that these challenges are a general property of the shapes. We further showed that NEG- t -SNE and Parametric UMAP are equivalent.

These insights into attraction–repulsion dynamics offer general-purpose tools for optimizing a wide range of dimensionality reduction algorithms. Beyond this, the close connection between dimensionality reduction and contrastive learning (Damrich and Hamprecht, 2021; Hu et al., 2023) suggests that our approach can also benefit representation learning more broadly. Taken together, our work aims to make embeddings and their interpretations more principled, consistent, and reliable across methods and applications, and to guide future research.

Software and Data

All datasets used in this paper are publicly available, spanning image and single-cell domains. The MNIST dataset is available at yann.lecun.com. Fashion-MNIST is available at github.com. Single-cell transcriptomes data is available at github.com.

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Appendix

In Appendix [A](#), we provide the necessary derivations for the attraction and repulsion shapes of UMAP. Following the discussion in the main text, we revisit the use of a constant learning rate in Appendix [B](#). In Appendix [C](#), we then discuss the implications of improved attraction and repulsion for large language model embeddings, illustrating that our findings extend to text-derived, high-dimensional representations. We formally define the Procrustes distance and matrix in Appendix [D.1](#). We report additional datasets for the random initialization experiment in Appendix [E](#) demonstrating that the conclusions hold across data domains. We extend the discussion of NEG-*t*-SNE and UMAP in Appendix [F](#). After that, in Appendix [G](#), we apply the same framework to a broad set of alternate dimensionality reduction methods. For completeness, we then discuss the construction of the high-dimensional graph in these methods in Appendix [H](#). In Appendix [I](#), we present detailed results of varying λ_a and λ_r (from Fig. [I](#)) for UMAP, NEG-*t*-SNE, and PaCMAP across different datasets. We show the effect of mixing and matching shapes from different algorithms in Appendix [J](#). Finally, we provide implementation details in Appendix [K](#).

A Proofs and Derivations

A.1 Proof of Proposition [4.1](#)

Proof. We subtract Eq. [\(7\)](#) from Eq. [\(8\)](#) and take a norm:

$$\|y_i^{t+1} - y_j^{t+1}\| = |1 + 2\lambda f_a| \|y_i^t - y_j^t\|. \quad (16)$$

This distance contracts as long as $|1 + 2\lambda f_a| < 1$, i.e., provided

$$-1 < \lambda f_a < 0, \quad (17)$$

□

A.2 Proof of Proposition [4.2](#)

Proof. We subtract Eq. [\(9\)](#) from Eq. [\(10\)](#) and take a norm:

$$\|y_i^{t+1} - y_j^{t+1}\| = |1 + \lambda f_r| \|y_i^t - y_j^t\|. \quad (18)$$

This distance increases when $|1 + \lambda f_r| > 1$, i.e.,

$$\lambda f_r < -2 \text{ or } f_r > 0. \quad (19)$$

From the definition of f_r , the latter suffices. □

A.3 Attraction Shape:

To keep the derivation general, we use a broad form of the low-dimensional affinity function, i.e., $q_{ij} = (\gamma + a\|y_i - y_j\|_2^{2b})^{-1}$, to derive the attraction shape. This single form covers multiple methods, reducing to UMAP for $\gamma = 1$ and to NEG-*t*-SNE for $\gamma = 2$. The attractive force is given by

$$\begin{aligned} \nabla_{y_i} \log q_{ij} &= -\nabla_{y_i} \log (\gamma + a(\|y_i - y_j\|_2^2)^b) \\ &= -\frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} (\gamma + a(\|y_i - y_j\|_2^2)^b) \\ &= -\frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} ab(\|y_i - y_j\|_2^2)^{b-1} \nabla_{y_i} \|y_i - y_j\|_2^2 \\ &= -\frac{2ab(\|y_i - y_j\|_2^2)^{b-1}}{\gamma + a(\|y_i - y_j\|_2^2)^b} (y_i - y_j). \end{aligned} \quad (20)$$

Defining $\zeta = \|y_i - y_j\|_2$, the first term gives the attraction shape as:

$$f_a(\zeta) = -\frac{2ab\zeta^{2(b-1)}}{\gamma + a\zeta^{2b}} \quad (21)$$

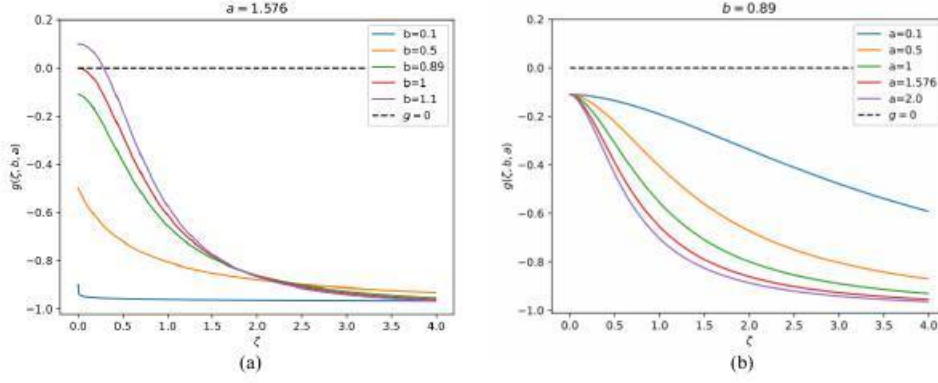


Figure 5: Values of $g(\zeta, b, a)$ for (a) a fixed at 1.576 and (b) b fixed at 0.89.

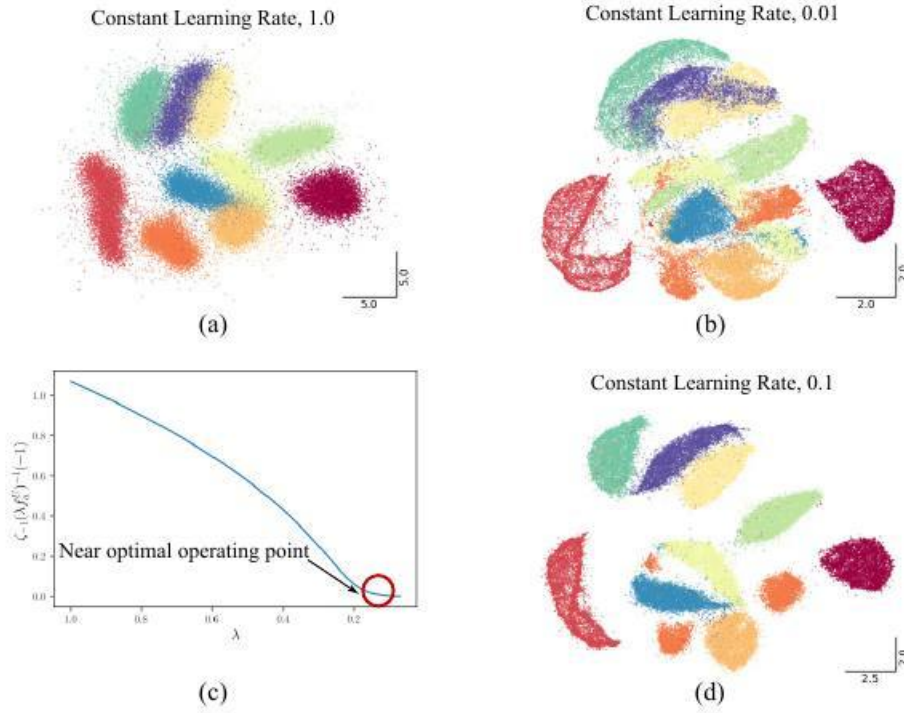


Figure 6: Effect of a constant learning rate on embeddings. (a) When the learning rate is too high ($\lambda = 1.0$), the embeddings are diffuse (owing to the high value of ζ_{-1}). (b) When the learning rate is too low ($\lambda = 0.01$), clusters fail to form (the strengths of attraction and repulsion are too low). (c) ζ_{-1} decreases nonlinearly as the learning rate decreases; the general goal is to reduce ζ_{-1} while maintaining effective levels of attraction and repulsion. (d) Distinct and compact clusters form at a constant, near-optimal learning rate $\lambda = 0.1$.

A.4 Condition for strictly increasing f_a^U :

Its behavior with distance can be characterized by computing the derivative of f_a^U :

$$\frac{df_a^U(\zeta)}{d\zeta} = -\frac{2ab\zeta^{2b-3}}{\gamma + a\zeta^{2b}} \left((b-1) - \frac{ab\zeta^{2b}}{\gamma + a\zeta^{2b}} \right). \quad (22)$$

This leads to a strictly increasing condition ($\frac{df_a^U}{d\zeta} > 0$),

$$g(\zeta, b, a) < 0, \quad (23)$$

where $g(\zeta, b, a) = b - 1 - \frac{ab\zeta^{2b}}{\gamma + a\zeta^{2b}}$. This inequality holds throughout the range $0 < b \leq 1$ (from the derivative and asymptotes of g). Figure 5 shows values of g for a range of b and a . As b increases above 1, the inequality (23) no longer holds.

A.5 Repulsion Shape:

The repulsive force, using the general form of the low-dimensional affinity, is given by

$$\begin{aligned} \nabla_{y_i} \log(1 - q_{ij}) &= \nabla_{y_i} \log \left[1 - \frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \right] \\ &= \frac{\gamma + a(\|y_i - y_j\|_2^2)^b}{(\gamma - 1) + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} \left[1 - \frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \right] \\ &= \frac{1}{\gamma - 1 + a(\|y_i - y_j\|_2^2)^b} \frac{ab(\|y_i - y_j\|_2^2)^{b-1}}{\gamma + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} \|y_i - y_j\|_2^2 \\ &= \frac{2ab(\|y_i - y_j\|_2^2)^{b-1}}{(\gamma - 1 + a(\|y_i - y_j\|_2^2)^b)(\gamma + a(\|y_i - y_j\|_2^2)^b)} (y_i - y_j). \end{aligned} \quad (24)$$

The first term gives the repulsion shape as:

$$f_r(\zeta = \|y_i - y_j\|_2) = \frac{2ab\zeta^{2(b-1)}}{(\gamma - 1 + a\zeta^{2b})(\gamma + a\zeta^{2b})}. \quad (25)$$

Generally, $f_r > 0$.

A.6 Loss functions due to modified attraction and repulsion shape:

The cost function of the attractive term with the modification in Eq. (13) is given by

$$- \int (f_a^U(\|y_i - y_j\|_2) - \beta\|y_i - y_j\|_2)(y_i - y_j) dy_i = -\log(q_{ij}) + \frac{\beta}{3}\|y_i - y_j\|_2^3, \quad (26)$$

whereas the repulsive term due to Eq. (15) is given by

$$- \int (f_r^U(\|y_i - y_j\|_2) + \epsilon)(y_i - y_j) dy_i = -\log(1 - q_{ij}) - \frac{\epsilon}{2}\|y_i - y_j\|_2^2. \quad (27)$$

In both cases, the additional term acts as a regularizer expressed simply through norms. Adding the term directly to the attraction and repulsion shapes, however, makes the role of each term transparent and easy to interpret in general: for the attractive term it increases far-sightedness, whereas for the repulsive term it adds a constant repulsive coefficient.

B More on UMAP's Learning Rate

As discussed in Sections 5 and 6 of the main text, UMAP is generally believed to require learning rate annealing (Fig 6). To explain this, in Section 5 we introduced the minimum distance for contraction (ζ_{-1}) and established the general principle that reducing this value through learning rate annealing yields compact clusters. Later, in Section 6 we compared the attraction shapes of UMAP (for $a = 1.0$ and $b = 1.0$) and NEG- t -SNE and explained that NEG- t -SNE tolerates a constant learning rate of 1.0 better than UMAP because its attraction shape resides within $[-1, 0]$ while UMAP's spans $[-2, 0]$. Following the same reasoning, we showed that UMAP can tolerate a constant learning rate of 0.5 once its attraction shape is constrained to $[-1, 0]$ (Figs. 11(f,h)).

Even so, the embeddings are still better when the learning rate anneals, and this holds across a wide range of parameters a and b . The reason is that the overarching goal is to eventually reduce ζ_{-1} to zero, which happens as the learning rate drops close to zero. Otherwise, the embedding becomes diffuse (Fig 6(a)). Conversely, if the learning rate is too low from the outset, the strengths of attraction and repulsion are too weak, and no clear clusters can form (Fig 6(b)). Examining the ζ_{-1} vs. λ curve (Fig 6(c)) reveals a near-optimal point at $\lambda = 0.1$, where ζ_{-1} is low while attraction and repulsion remain appreciable. Setting the constant learning rate to 0.1 yields compact clusters with clear boundaries (Fig 6(d)).

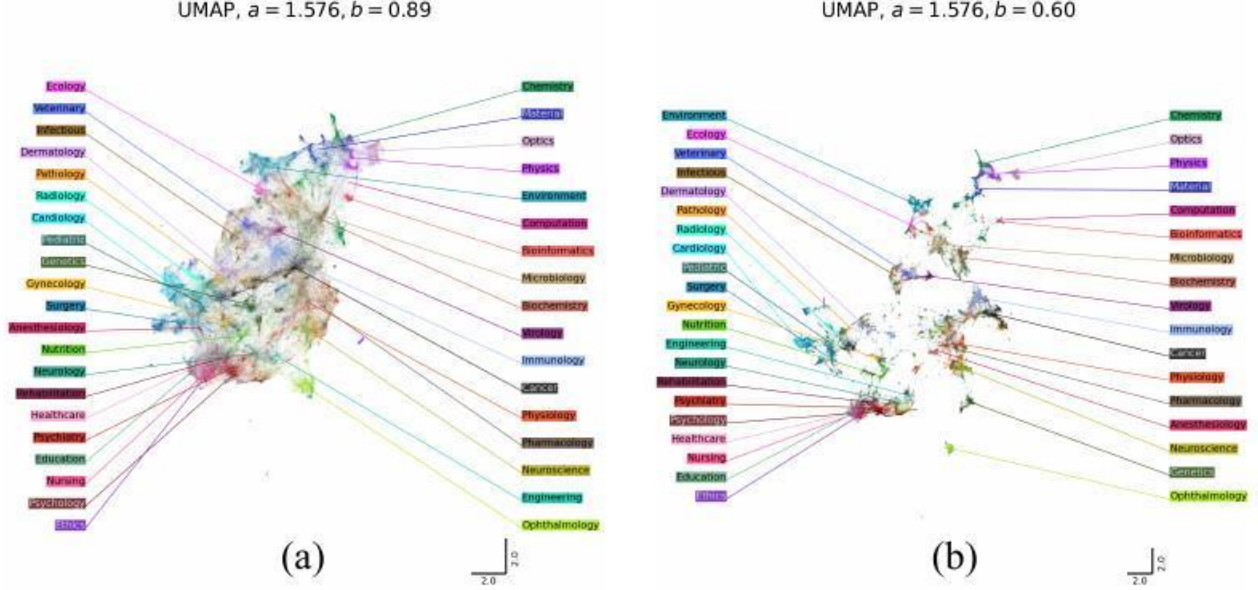


Figure 7: Visualization of 300K abstracts from the PubMed dataset, a large-scale text-embedding setting. (a) Default UMAP ($a = 1.576, b = 0.89$) yields overlapping labels in the mapping, whereas (b) UMAP with improved repulsion and attraction ($a = 1.576, b = 0.60$) yields better cluster formation and non-overlapping labels. Both mappings are labeled for ease of interpretation.

C Large Language Model Embedding

Increased attraction and repulsion often produce better embeddings, and this benefit extends well beyond image data to high-dimensional language model representations. To demonstrate this on a large-scale text-embedding task, we randomly selected 300,000 feature samples produced by PubMedBERT (Gu et al. 2021) with known labels from the PubMed dataset (González-Márquez et al. 2024). Default UMAP yields a crowded structure in which labels (colors) overlap (Fig. 7a). Without labels, one might struggle to interpret these embeddings, as explicit clusters are absent. By adjusting b to 0.6 (as discussed in Section 5.2), however, we increase repulsion while simultaneously promoting cluster formation through improved attraction. The resulting mapping is explorable even without explicit labeling (Fig. 7b). The improvement can be quantified by the k -NN accuracy ($k = 10$). On a hold-out set of 10,000 points, reducing b to 0.6 raises the k -NN accuracy from 49.2% to 55.48%, confirming that the same shape-based principles that help image embeddings also carry over to language model embeddings.

D Metrics

D.1 Procrustes Distance and Procrustes Matrix

The Procrustes distance (Gower 1975) measures the similarity between two point clouds $\{x\}$ and $\{y\}$ under linear transformations, namely translation, scaling, and rotation. Operationally, we hold the first set fixed and vary the second until the two are in maximum alignment. Let $\{y'\}$ be the transformation of $\{y\}$ that achieves this objective. The Procrustes distance is then given by

$$pd(\{x\}, \{y\}) = \sqrt{\sum_k (x_k - y'_k)^2}. \quad (28)$$

The Procrustes distance is a linear measure that has proven useful across a wide variety of settings (McInnes et al. 2018; Islam and Fleischer 2022; Kotlov et al. 2024).

Here, we use the Procrustes distance to measure the consistency of embeddings under random initialization. Let $\{x\}_p$ be a reference embedding (PCA initialization in our experiments), and let $X_r = \{\{x\}_i | i = 1, 2, 3, \dots, N\}$ be a set of N embeddings obtained from random initialization. The similarity of the embeddings can be quantified by the mean and standard deviation of the strictly lower-triangular values of the matrix P (reported as mean $\pm$ std in Figs. 2 9).

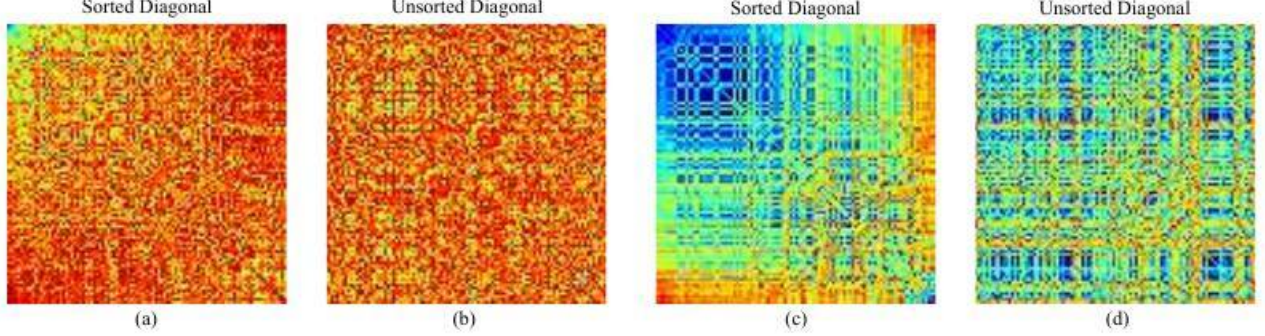


Figure 8: Effect of sorting the diagonal of the Procrustes matrix on its visualization. (a) Procrustes matrix reproduced from [2]f) for the default UMAP attraction shape, with the diagonal sorted by p_d of a sample embedding with PCA initialization. (b) The same data as in (a) but with the diagonal unsorted (or randomly sorted). (c) Procrustes matrix reproduced from [2]g) for the modified attraction shape, with the diagonal sorted by p_d of a sample embedding with PCA initialization. (d) The same data as in (c) but with the diagonal unsorted.

and [10], with

$$\text{mean} = \frac{2}{N(N-1)} \sum_{i,j(i>j)} P_{i,j}, \text{ and } \text{std} = \sqrt{\sum_{i,j(i>j)} \frac{2(P_{i,j} - \text{mean})^2}{N(N-1)}} \quad (29)$$

The indexes of X_r can be sorted such that $p_d(\{x\}_i, \{x\}_p) \leq p_d(\{x\}_{i+1}, \{x\}_p)$, so that the diagonal values of the Procrustes matrix are given by

$$P_{i,i} = p_d(\{x\}_i, \{x\}_p) \quad (30)$$

and the off-diagonal values are given by

$$P_{i,j} = p_d(\{x\}_i, \{x\}_j). \quad (31)$$

Numerically, sorting the diagonal adds little value. Visually, however, the ordering reveals the underlying self-similarity of the embeddings. For example, in Fig. [8]a), the sorted diagonal shows that similar embeddings cluster in the upper-left region of the matrix. With a modified attraction shape, the number of mutually similar embeddings increases (as shown by the larger blue region in Fig. [8]c)), indicating a metastable point in the embedding algorithm. When the diagonal is left unsorted, by contrast, this region disappears and any sense of similar embeddings is lost (Fig. [8]b,d)).

D.2 Trustworthiness

The trustworthiness metric [Venna and Kaski, 2001] is a general-purpose measure that quantifies how well local neighborhoods are preserved after dimensionality reduction:

$$T = 1 - \frac{2}{nk(2n - 3k - 1)} \sum_{i=1}^n \sum_{y_j \in \text{KNN}(y_i, k)} \max(0, r(i, j) - k) \quad (32)$$

where $\text{KNN}(y_i, k)$ is the k -NN graph in the embedding space and $r(i, j)$ is the rank of x_j in the high-dimensional k -NN graph. In practice, k is commonly set to 5, assessing preservation of each point's five nearest neighbors. For computational efficiency, whenever we report trustworthiness we randomly sampled 10,000 indices, and we used the same indices when comparing different embeddings to ensure a fair comparison.

D.3 Silhouette Score

Although the silhouette score [Rousseeuw, 1987] is designed to assess clustering algorithms, we repurpose it here to evaluate label separation in the embeddings, i.e., how well the ground-truth labels have been clustered. The rationale is that embedding algorithms naturally produce clusters and should separate the labels as much as possible. For a point y_i in a point cloud $\{y\}$, let a_i be the mean distance from y_i to other points sharing its label, and let b_i be the minimal mean distance from y_i to points in any other label. The pointwise silhouette is then given by

$$s_i = \frac{b_i - a_i}{\max\{a_i, b_i\}} \quad (33)$$

This value lies within $[-1, 1]$. A value close to 1 indicates that y_i fits well within its own label cluster, a value near 0 suggests a boundary point, and a value close to -1 indicates failed label separation. The overall silhouette score is

$$\text{SIL} = \frac{1}{N} \sum s_i. \quad (34)$$

We computed the silhouette score over the entire embedding (without random sampling) using Euclidean distances.

E Effect of Random Initialization in Additional Datasets

In the main text (Section 5.1), we presented results on the MNIST dataset. To establish that our findings are not tied to a single dataset or data type, we repeat the same experiment here on the Fashion-MNIST (FMNIST) (Xiao et al., 2017) image data and the single-cell transcriptomes (Macosko et al., 2015) biological data (Figs. 9 and 10 respectively). The main conclusion holds unchanged across these distinct domains: modified and composite attraction shapes, such as those that increase attraction at large distances, substantially improve the consistency of reconstruction.

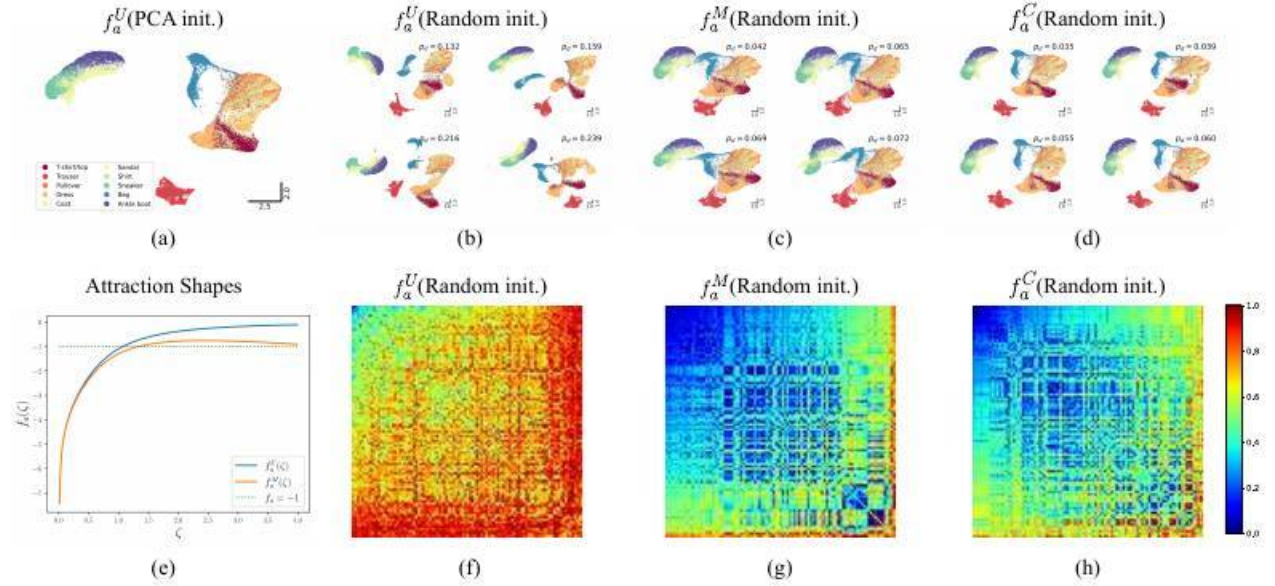


Figure 9: Effect of random UMAP initialization on different attraction shapes for the FMNIST image data, confirming that the MNIST findings generalize. (a) Mapping using PCA as a reference. (b-d) Four mappings with the lowest Procrustes distance (p_d) from the embedding in (a) for (b) default, (c) modified, and (d) composite attraction shapes. (e) Default UMAP and modified attraction shapes. (f-h) Procrustes matrix obtained from 100 runs of (f) default (0.72 ± 0.15), (g) modified (0.38 ± 0.19), and (h) composite (0.43 ± 0.18) attraction shapes.

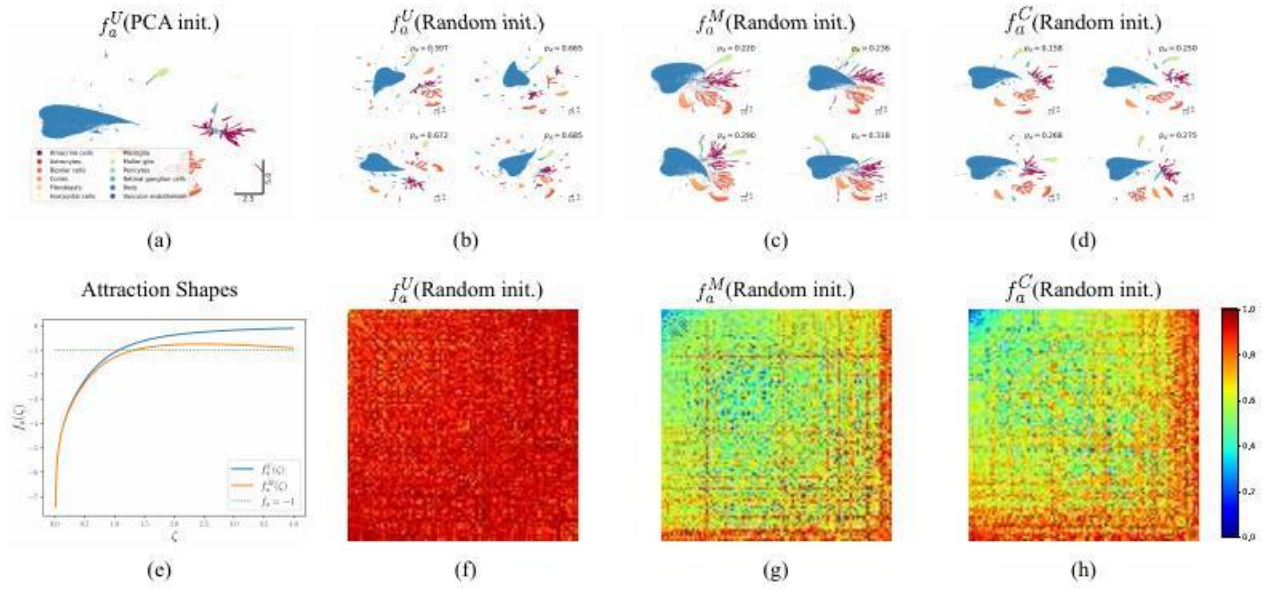


Figure 10: Effect of random UMAP initialization on different attraction shapes for the single-cell transcriptomes biological data, demonstrating that the conclusions extend to a very different data domain. (a) Mapping using PCA as a reference. (b-d) Four mappings with the lowest Procrustes distance (p_d) from the embedding in (a) for (b) default, (c) modified, and (d) composite attraction shapes. (e) Default UMAP and modified attraction shapes. (f-h) Procrustes matrix obtained from 100 runs of (f) default (0.91 ± 0.06), (g) modified (0.61 ± 0.13), and (h) composite (0.64 ± 0.15) attraction shapes.

F More on NEG- t -SNE

F.1 Comparison to UMAP

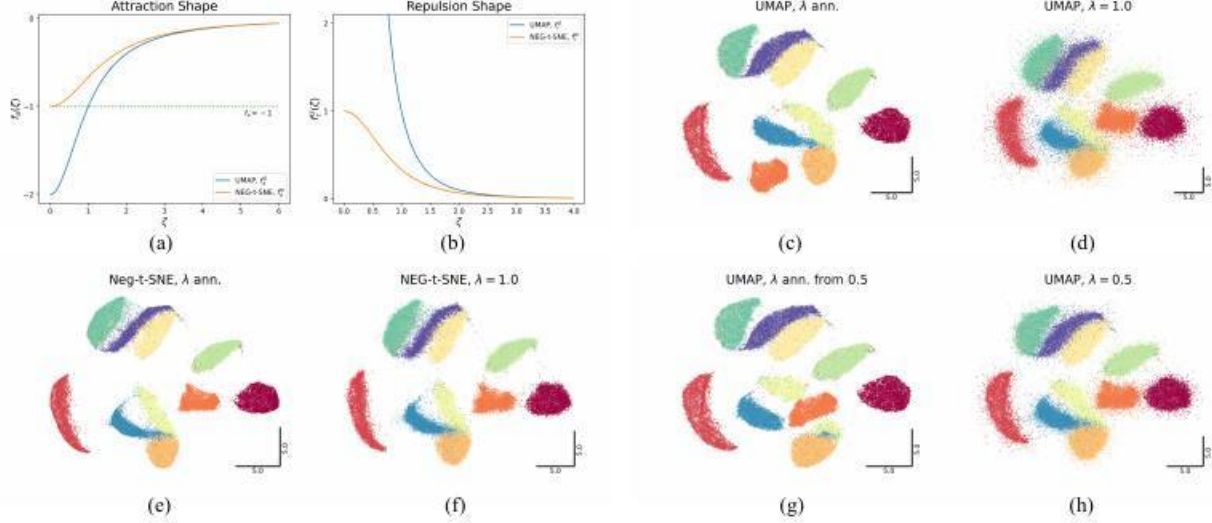


Figure 11: Sensitivity of UMAP and NEG- t -SNE to the learning rate, illustrated on the MNIST dataset. (a) Attraction and (b) repulsion shapes for UMAP ($a = 1$, $b = 1$) and NEG- t -SNE. (c,d) UMAP is highly sensitive to the learning rate λ , since $f_a^U < -1$ as the separation distance ζ decreases; without annealing, the clusters become fuzzy. (e,f) NEG- t -SNE is less sensitive to λ because $f_a^N \in [-1, 0]$ always, so the clusters remain less fuzzy even without annealing. (g,h) Confining f_a^U to $[-1, 0]$ by setting $\lambda = 0.5$ likewise reduces sensitivity to λ .

Figure 11 shows the shapes of UMAP and NEG- t -SNE, along with several MNIST embeddings. When the learning rate is annealed, UMAP and NEG- t -SNE produce similar output (Figs. 11(c,e)). At a constant learning rate of 1, however, UMAP yields a fuzzy structure while NEG- t -SNE yields much cleaner boundaries (Fig. 11(d,f)). The discussion in Section 6 implies, as a general prediction, that constraining f_a^U within $[-1, 0]$ should produce less fuzzy clusters for fixed λ . We saw the same effect earlier in Fig. 11(h,i), where UMAP gave better embedding and clustering when the attraction learning rate (λ_a) was $\lesssim 0.5$. A simple way to achieve this is to initialize λ to 0.5, which satisfies Proposition 4.1 for all ζ . The resulting embeddings, shown in Figs. 11(g) and (h), confirm that the clusters resemble those of NEG- t -SNE, and that for a constant $\lambda = 0.5$ the clusters are less fuzzy than before, with sharper boundaries as predicted (Fig. 11(d,h)). A few points still lie outside the clusters, owing to the characteristics of UMAP’s repulsion shape, which NEG- t -SNE resolves.

Next, we can introduce the parameters a and b into NEG- t -SNE (essentially the formulation of Parametric UMAP; see Proposition 6.1). The affinity function becomes $q_{ij}^N = 1/(2 + a\zeta^{2b})$, and the attraction and repulsion shapes become $f_a^N = -\frac{2ab\zeta^{2(b-1)}}{2+a\zeta^{2b}}$, and $f_r^N = \frac{2ab\zeta^{2(b-1)}}{(1+a\zeta^{2b})(2+a\zeta^{2b})}$, respectively. For $0 < b < 1$, both shapes become unbounded as $\zeta \rightarrow 0$. NEG- t -SNE therefore faces the same numerical challenges as UMAP once a and b vary, so the corresponding limitations carry over quite generally. One notable distinction is that, relative to UMAP, the attraction shape attains a lower minimum distance for contraction (ζ_{-1}). While this can enhance cluster formation, it also approaches zero faster (increasing near-sightedness at larger distances), potentially diminishing its effectiveness for attraction over longer distances.

F.2 Comparison to Parametric UMAP

Parametric UMAP was initially trained with the original UMAP objective (Sainburg et al., 2021), but later work adopted a numerically stable, log-sigmoid-based modified cross-entropy loss (Shi et al., 2023). This modification makes Parametric UMAP and NEG- t -SNE (Damrich et al., 2023) equivalent (Proposition 6.1), a correspondence that unifies two methods often treated as distinct. We establish the equivalence below.

Proof of Proposition 6.1

Proof. The kernel function under this modification becomes

$$q_{ij}^P = -\log(1 + a\|y_i - y_j\|_2^{2b}). \quad (35)$$

The attractive term is

$$-\text{logsigmoid}(q_{ij}^P) = -\log\left(\frac{1}{1 + \exp(-q_{ij}^P)}\right) \quad (36)$$

$$= -\log\left(\frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (37)$$

$$= -\log(q_{ij}^N). \quad (38)$$

And the repulsive term is

$$\text{logsidmoid}(q_{ij}^P) - q_{ij}^P = \log\left(\frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) + \log(1 + a\|y_i - y_j\|_2^{2b}) \quad (39)$$

$$= \log\left(\frac{1 + a\|y_i - y_j\|_2^{2b}}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (40)$$

$$= \log\left(1 - \frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (41)$$

$$= \log(1 - q_{ij}^N). \quad (42)$$

Both these are NEG-t-SNE with explicit parameters a and b (while in Neg-t-SNE these are set to 1). $\square$

G Alternate Dimensionality Reduction Algorithms

To underscore the breadth of our framework, we now apply it to a range of alternative algorithms. Each of them uses the same kernel function as UMAP (with $a = 1$ and $b = 1$ in their low-dimensional weight):

$$q_{ij} = \frac{1}{1 + \|y_i - y_j\|_2^2}. \quad (43)$$

In this section, we first discuss the TriMap (Amid and Warmuth 2019) algorithm. Although it relies on triplets rather than pairwise interactions, it serves as a useful primer for analyzing attraction and repulsion in cases somewhat more involved than UMAP. We then turn to Pairwise Controlled Manifold Approximation (PaCMAP) (Wang et al. 2021) and its extension, Pairwise Controlled Manifold Approximation with Local Adjusted Graph (LocalMAP) (Wang et al. 2025), which adapt TriMap’s loss function to pairwise interactions. Next we tackle t -SNE (Van der Maaten and Hinton, 2008), provide a short note on SNE (Hinton and Roweis 2002) (which uses an alternate kernel function), and close the section by briefly discussing multidimensional scaling (Borg and Groenen 2007). Taken together, these cases show that the attraction-repulsion shape framework accommodates methods with markedly different loss structures.

G.1 TriMap

TriMap (Amid and Warmuth 2019) optimizes low-dimensional using a triplet loss

$$\mathcal{L}^T = \sum_{(i,j,k)} w_{ijk} \frac{1}{1 + \frac{q_{ij}}{q_{ik}}}, \quad (44)$$

where w_{ijk} is the weight of the triplet (y_i, y_j, y_k) , y_j lies in the k -nearest neighbor set of y_i in high dimension, and y_k is a far-away point. Minimizing this loss is expected to make y_i and y_j attract each other while y_i and y_k repel each other. The update equations are

$$y_i^{t+1} = y_i^t + \lambda f_a^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_j^t) + \lambda f_r^T(\zeta_1, \zeta_2)(y_i^t - y_k^t), \quad (45)$$

$$y_j^{t+1} = y_j^t - \lambda f_a^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_j^t), \quad (46)$$

$$y_k^{t+1} = y_k^t - \lambda f_r^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_k^t), \quad (47)$$

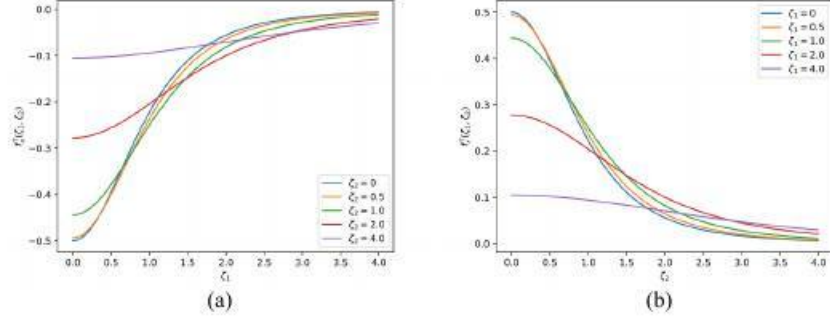


Figure 12: (a) Attraction shapes of TriMap across different values of ζ_2 and (b) repulsion shapes of TriMap across different values of ζ_1 .

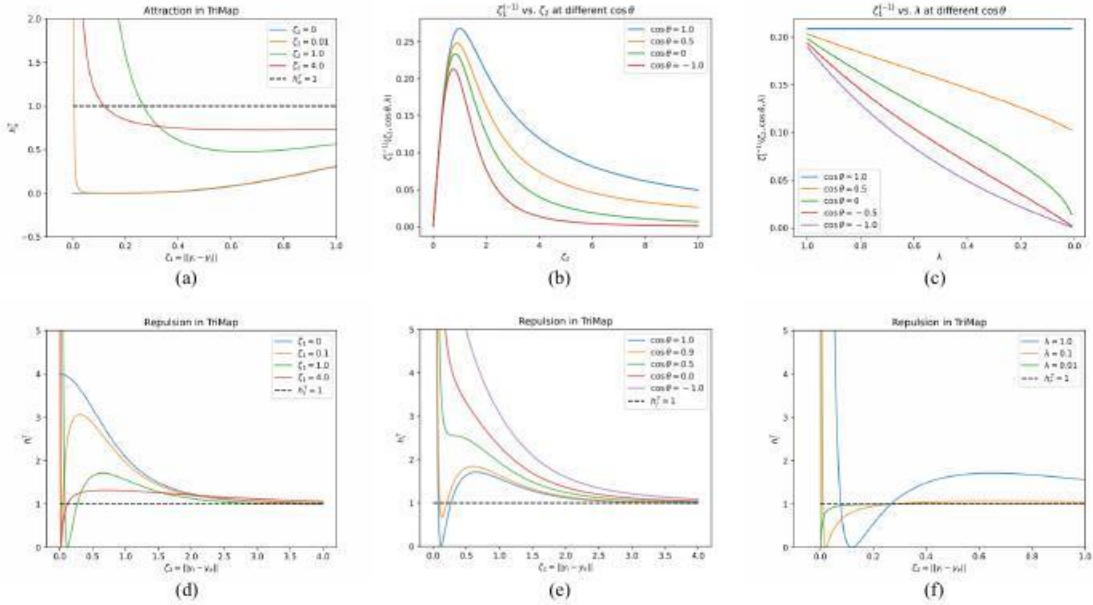


Figure 13: Attraction and repulsion behavior in TriMap. (a) h_α^T vs. ζ_1 for different ζ_2 . Values below the dotted line indicate attraction. As in UMAP, TriMap exhibits both attraction and repulsion for nearest neighbors (y_i, y_j) . (b,c) Unlike UMAP, the minimum distance for contraction ($\zeta_1^{(-1)}$) varies because of its dependence on (b) ζ_2 and (c) λ ; the function $\cos \theta$ regulates the range of these values. (d-f) Repulsion in TriMap as we vary (d) ζ_1 , (e) $\cos \theta$, and (f) λ . Values above (below) the dotted line indicate repulsion (attraction). Whereas UMAP's repulsion force yields only repulsion, TriMap's can yield both attraction and repulsion. Unless otherwise labeled, $\zeta_1 = 1.0$, $\zeta_2 = 0.5$, $\cos \theta = 1.0$, and $\lambda = 1.0$.

where $\zeta_1 = \|y_i - y_j\|_2$ is the distance between nearest neighbors, $\zeta_2 = \|y_i - y_k\|_2$ is the distance to the faraway point, and f_a^T and f_r^T are the attraction and repulsion shapes of TriMap, respectively. Unlike UMAP, the attractive and repulsive components here are non-separable, and the shapes depend on two distance measures (making them two-dimensional) — a more general situation that our framework still accommodates. The functional form of the attraction shape is

$$f_a^T(\zeta_1, \zeta_2) = -\frac{2(1 + \zeta_2^2)}{(2 + \zeta_1^2 + \zeta_2^2)^2}, \quad (48)$$

and the repulsion shape is

$$f_r^T(\zeta_1, \zeta_2) = \frac{2(1 + \zeta_1^2)}{(2 + \zeta_1^2 + \zeta_2^2)^2}. \quad (49)$$

The attraction and repulsion shapes of TriMap (Fig. 12) follow trends similar to those of UMAP. However, the minimum value of the repulsion shape is -0.5 , so, unlike UMAP, no position flipping occurs in TriMap due to attraction alone. Because Eqs. (45)-(47) do not decouple the attractive and repulsive terms, Propositions 4.1 and 4.2 do not apply directly; this motivates a more general condition. Focusing on attraction first, we show

Proposition G.1. *Update equations (45)-(47) provide a contraction if*

$$h_a^T(\zeta_1, \zeta_2, \theta, \lambda) < 1, \quad (50)$$

where $h_a^T(\zeta_1, \zeta_2, \theta, \lambda) = (1 + 2\lambda f_a^T)^2 + 2(1 + 2\lambda f_a^T)\lambda f_r^T \frac{\zeta_2}{\zeta_1} \cos \theta + (\lambda f_r^T)^2 \frac{\zeta_2^2}{\zeta_1^2}$ and θ is the angle between the vectors $(y_i - y_j)$ and $(y_i - y_k)$, i.e., $\cos \theta = \frac{(y_i^t - y_j^t)^T (y_i^t - y_k^t)}{\|y_i^t - y_j^t\|_2 \|y_i^t - y_k^t\|_2}$.

Proof. We require

$$\|y_i^{t+1} - y_j^{t+1}\|_2^2 < \|y_i^t - y_j^t\|_2^2. \quad (51)$$

From Eq. (45) and (46): $y_i^{t+1} - y_j^{t+1} = (1 + 2\lambda f_a^T)(y_i^t - y_j^t) + \lambda f_r^T (y_i^t - y_k^t)$. Putting this value in Eq. (51), we obtain the desired inequality. $\square$

Inequality (50) depends on ζ_1 , ζ_2 , $\cos \theta$, and λ . In particular, the value $\zeta_1 = \|y_i - y_j\|_2$, where we seek contraction, is coupled with additional variables. Figure 13(a) shows the attraction behavior for various values of ζ_2 with $\cos \theta = 1$ and $\lambda = 1$. Values below the dotted line indicate attraction (contraction of distance ζ_1), whereas values above it indicate repulsion (expansion of distance ζ_1). The point where the dotted line meets h_a^T gives the minimum distance for contraction ($\zeta_1^{(-1)}$), the natural analogue of ζ_{-1} in UMAP, which we define as

$$\zeta_1^{(-1)}(\zeta_2, \theta, \lambda) = \arg \min_{\zeta_1} |h_a^T(\zeta_1, \zeta_2, \theta, \lambda) - 1|, \quad (52)$$

$$\text{s.t. } \zeta_1 \geq 0. \quad (53)$$

$\zeta_1^{(-1)}$ is finite and > 0 in most cases (Fig. 13(b,c)). Consequently, TriMap can exhibit behavior similar to UMAP and thus require learning rate annealing or a comparable mechanism (in the TriMap implementation, the authors use the delta-bar-delta (Jacobs 1988) method under appropriate initialization) — another instance of the same general pattern.

Proposition G.2. *Update equations (45)-(47) provide expansion if*

$$h_r^T(\zeta_1, \zeta_2, \theta, \lambda) > 1, \quad (54)$$

where $h_r^T(\zeta_1, \zeta_2, \theta, \lambda) = (1 + 2\lambda f_r^T)^2 + 2\lambda f_a^T (1 + \lambda f_r^T) \frac{\zeta_1}{\zeta_2} \cos \theta + (\lambda f_a^T)^2 \frac{\zeta_1^2}{\zeta_2^2}$.

Proof. We require

$$\|y_i^{t+1} - y_k^{t+1}\|_2^2 > \|y_i^t - y_k^t\|_2^2. \quad (55)$$

From Eq. (45) and (47): $y_i^{t+1} - y_k^{t+1} = (1 + 2\lambda f_r^T)(y_i^t - y_k^t) + \lambda f_a^T (y_i^t - y_j^t)$. Putting this value in Eq. (55) we obtain inequality (54). $\square$

Inequality (54) likewise depends on ζ_1 , ζ_2 , $\cos \theta$, and λ . Here, we are interested in the expansion of $\zeta_2 = \|y_i - y_k\|_2$. Figures 13(d-f) show the repulsion behavior as we vary the other quantities. Values above the dotted line indicate repulsion (expansion of distance), while values below it indicate attraction. The striking difference from UMAP is that repulsion in TriMap can cause contraction rather than expansion. Since this anomaly arises only at small distances, it can be avoided through appropriate initialization and choice of triplets.

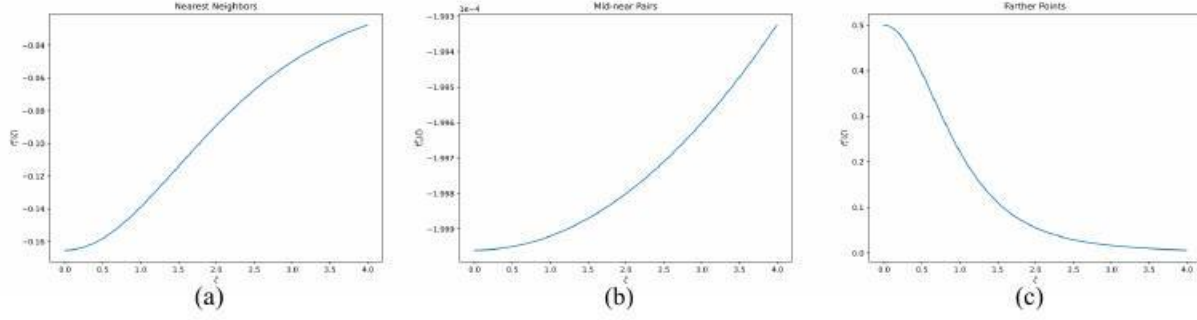


Figure 14: Attraction and repulsion shapes of PaCMAP. (a,b) Attraction shapes for (a) nearest-neighbor and (b) mid-near points (note that the values are on the order of 10^{-4}). (c) Repulsion shape for farther pairs. Here $\lambda = 1$ throughout.

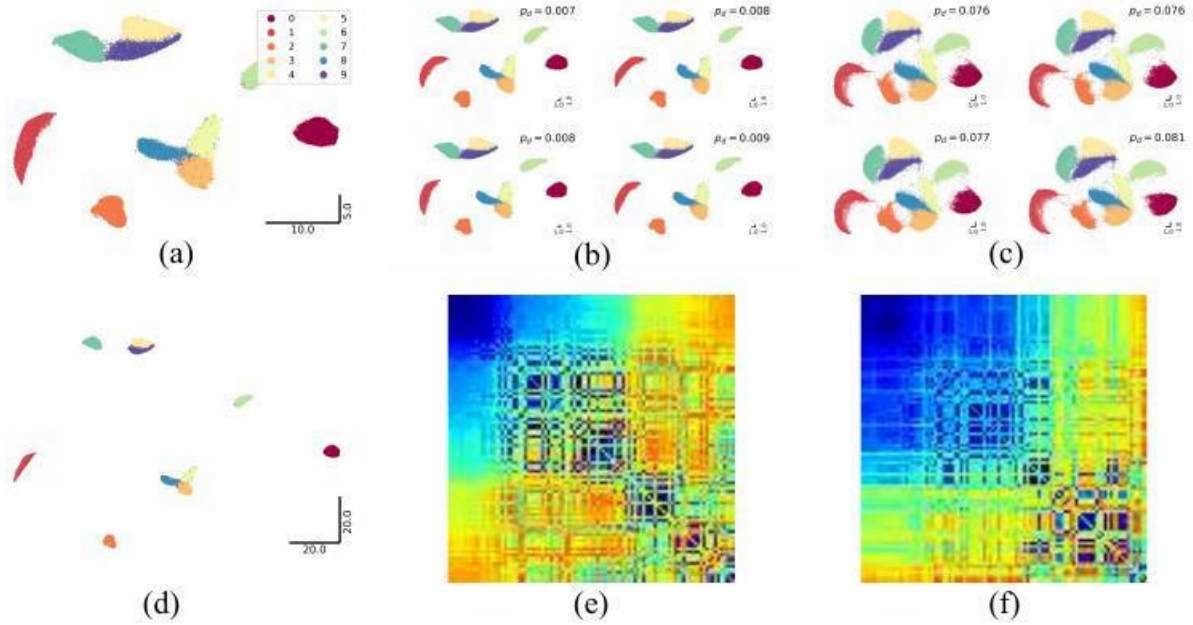


Figure 15: PaCMAP behavior under different conditions. (a) PaCMAP of MNIST data with PCA initialization. (b,c) Four samples that best match (a), obtained by (b) original PaCMAP and (c) PaCMAP with a modified attraction shape, $f_a^M(\zeta) = f_a^P(\zeta) - 0.001\zeta$, when randomly initialized. (d) PaCMAP of MNIST with a modified repulsion shape $f_r^M = f_r^P + 0.00005$. (e,f) Procrustes matrix for (e) original PaCMAP (0.51 ± 0.22) and (f) PaCMAP with a modified attraction shape (0.43 ± 0.22). As in UMAP, increased attraction at farther distances improves consistency, while increased repulsion produces smaller clusters and larger inter-cluster distances — confirming that the same shape-based levers act consistently across methods.

G.2 Pairwise Controlled Manifold Approximation (PaCMAP)

PaCMAP (Wang et al., 2021) optimizes low-dimensional embedding at different scales. The loss function is

$$\mathcal{L}^P = w_{NB} \sum_{(i,j) \in NN} \frac{1}{1 + 10q_{ij}} + w_{MN} \sum_{(i,k) \in MN} \frac{1}{1 + 10000q_{ik}} + w_{FP} \sum_{(i,l) \in FP} \frac{1}{1 + \frac{1}{q_{il}}}, \quad (56)$$

where w_{NB} , w_{MN} , and w_{FP} are the weights of the nearest neighbor (NN) pairs, mid-near (MN) pairs, and further pairs (FP), respectively (details in Appendix H). The first two terms provide attraction, while the last term provides repulsion. A closer look reveals that this loss is a modified form of TriMap’s triplet loss: for the attractive terms, it replaces TriMap’s affinity for distant points with constant terms ($1/10$ for nearest neighbors, $1/10000$ for mid-near pairs, and 1 for farther points). The loss is therefore separable into three terms and decouples the update equations. The

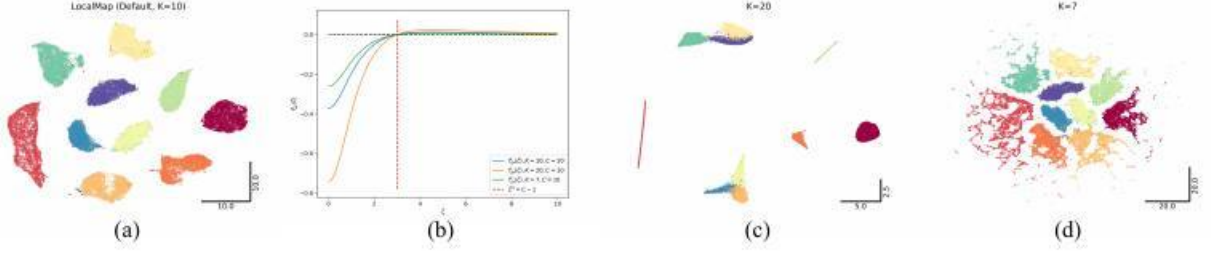


Figure 16: Behavior of LocalMAP on MNIST data. (a) Default embedding. (b) Attraction-repulsion shape for nearest neighbors. A value below (above) the dotted line indicates attraction (repulsion). The transition from attraction to repulsion occurs as ζ increases past $\sqrt{C}-1$. Following the LocalMAP implementation, we used $C = 10$. (c) When K is large ($= 20$), repulsion dominates and clusters become compact. (d) When K is small ($= 7$), attraction dominates and clusters break apart.

update equations for the nearest neighbor term are

$$y_i^{t+1} = y_i^t + \lambda f_a^P(\zeta_1^t)(y_i^t - y_j^t), \quad (57)$$

$$y_j^{t+1} = y_j^t - \lambda f_a^P(\zeta_1^t)(y_i^t - y_j^t), \quad (58)$$

of the mid-near pairs are

$$y_i^{t+1} = y_i^t + \lambda f_m^P(\zeta_2^t)(y_i^t - y_k^t), \quad (59)$$

$$y_k^{t+1} = y_k^t - \lambda f_m^P(\zeta_2^t)(y_i^t - y_k^t), \quad (60)$$

and of the farthest pairs are

$$y_i^{t+1} = y_i^t + \lambda f_r^P(\zeta_3^t)(y_i^t - y_l^t), \quad (61)$$

$$y_l^{t+1} = y_l^t - \lambda f_r^P(\zeta_3^t)(y_i^t - y_l^t), \quad (62)$$

where $\zeta_1 = \|y_i - y_j\|_2$, $\zeta_2 = \|y_i - y_k\|_2$, $\zeta_3 = \|y_i - y_l\|_2$ are distances, f_a^P and f_m^P are attraction shapes for nearest neighbors and mid-near pairs, respectively, and f_r^P is the repulsion shape for the farthest pairs. Correspondingly, the functional forms of the shapes are

$$f_a^P(\zeta) = -\frac{20}{(11 + \zeta^2)^2}, \quad (63)$$

$$f_m^P(\zeta) = -\frac{20000}{(10001 + \zeta^2)^2}, \quad (64)$$

$$f_r^P(\zeta) = \frac{2}{(2 + \zeta^2)^2}. \quad (65)$$

f_a^P and f_m^P follow Proposition 4.1 and f_r follows Proposition 4.2 (Fig. 14). The attraction is quite weak compared to UMAP, yet it suffices across a wide range of learning rates (modulated by the Adam algorithm (Kingma and Ba 2015)); with $w_{NB} = 3$, the maximum attraction always stays below 0.5, preventing any flips during the attraction update. Typically, PaCMAP initializes the embedding within a small sphere in the low dimension (e.g., in (Wang et al. 2021), the initialization is often on the order of 10^{-3}) and relies on repulsion to separate the individual clusters. Overall, it largely recovers UMAP’s clustering properties (especially for MNIST) with improved ordering. Its consistency under random initialization is better than UMAP’s (Fig. 15 b,e) and can be improved further with a modified attraction (Fig. 15 c,f). Increasing repulsion by adding a small value to the repulsion shape increases the compactness of the embedding (by widening inter-cluster distances) — the same behavior we observed for UMAP, now reproduced for PaCMAP.

G.3 Pairwise Controlled Manifold Approximation with Local Adjusted Graph (LocalMAP)

LocalMAP (Wang et al. 2025) is a refinement of the PaCMAP algorithm. One of its defining features is the separation of all 10 clusters of the MNIST data (Fig. 16 a)), with behavior similar to that seen in Figs. 3 g,h,j,k) and Figs. 4 a,b). Here, we again examine how the interplay of attractive and repulsive forces shapes the compactness and connectedness of clusters, further evidencing that this interplay is a general organizing principle.

LocalMAP performs PaCMAP and then does additional optimization on the attraction to decouple some clusters. To this end, it minimizes the following loss function

$$\mathcal{L}^L = \sum_{(i,j) \in NN} \frac{K}{\frac{1}{\sqrt{q_{ij}}} + C\sqrt{q_{ij}}} + \sum_{(i,l) \in FP} \frac{1}{1 + \frac{1}{q_{il}}}. \quad (66)$$

The first term amalgamates the attractive and repulsive character of the triplet loss, both acting on the same pair. In one regime this function causes attraction, and in the other it causes repulsion. The second term is identical to the one in PaCMAP, differing only in that the algorithm resamples further pairs every few iterations. We therefore analyze only the first term, involving nearest neighbors. The update equations are

$$y_i^{t+1} = y_i^t + \lambda f_{ar}^L(\zeta_1^t)(y_i^t - y_j^t), \quad (67)$$

$$y_j^{t+1} = y_j^t - \lambda f_{ar}^L(\zeta_1^t)(y_i^t - y_j^t), \quad (68)$$

where f_{ar}^L is the attraction-repulsion shape given by

$$f_{ar}^L(\zeta) = -\frac{K(C-1-\zeta^2)}{2\sqrt{1+\zeta^2}(1+C+\zeta^2)^2}. \quad (69)$$

The update yields contraction as long as $\zeta^2 < C-1$ and $-1 < \lambda f_{ar}^L < 0$ (from Proposition 4.1). When $\zeta^2 > C-1$, we have $f_{ar} > 0$, and by Proposition 4.2 the update equations yield expansion. The values of K and C jointly determine whether attraction or repulsion dominates the dynamics. In the LocalMAP implementation, both C and K are set to 10 (Fig. 16b)), and the strength of attraction is higher than in PaCMAP (Fig. 14a)). As a result, when $\zeta > 3$ the nearest neighbors experience repulsion, causing pairs that bridge two clusters to separate. The parameter K , acting as a scaling factor for the forces, regulates this separation.

Using Proposition 4.1 the condition $\lambda f_{ar}^L(0) \geq -\frac{1}{2}$ gives the maximum value of K , and hence the maximum attraction attainable by f_{ar}^L , without flipping the placements of the pairs. (We wish to avoid flipping pairs during LocalMAP optimization so as to preserve the ordering inherited from PaCMAP; otherwise, cluster separation may be inhibited.) This simplifies (69) to $K \leq \frac{\lambda(1+C)^2}{C-1}$, which for $C = 10$ and $\lambda = 1$ gives $K \lesssim 13.44$. At higher values $K \gtrsim 13.44$, the repulsive forces dominate and the clusters become more compact, but the objective of LocalMAP fails (the bridge between clusters persists in Fig. 16c) for $K = 20$). Conversely, as K decreases, attractive forces dominate (because repulsive forces are too weak), and the embedding shows existing clusters breaking apart (Fig. 16d) for $K = 7$, mirroring Fig. 4b)).

G.4 t -distributed Stochastic Neighbor Embedding (t -SNE)

t -SNE (Van der Maaten and Hinton [2008]) optimizes pairwise distances. The loss function is

$$\mathcal{L} = -\sum_{i,j} w_{i,j} \log \left(\frac{q_{i,j}}{\sum_{k \neq l} q_{k,l}} \right), \quad (70)$$

where $w_{i,j}$ is the weight of the pair. The original implementation of t -SNE considers all pairs, not just nearest neighbors. This loss function decomposes into attractive and repulsive forces as

$$\mathcal{L} = \sum_{i,j} \left[-w_{i,j} \log(q_{i,j}) + w_{i,j} \log \left(\sum_{k \neq l} q_{k,l} \right) \right]. \quad (71)$$

As before, the first term provides the attractive forces and the second term the repulsive forces. While the attractive term is identical to that of UMAP (with $a = 1$ and $b = 1$) and is easy to compute, the repulsive term is coupled across every pair and is therefore very costly. Applying the same principles we used for UMAP, we can write the update equations of t -SNE, which shows that the shape framework transfers even to this normalized, all-pairs setting. Since the original t -SNE does not rely on the nearest neighbor graph, the weight $w_{i,j}$, computed over all pairs, plays an important role in the update equations. The attractive update equations are

$$y_i^{t+1} = y_i^t + \lambda w_{i,j} f_a^{t-SNE}(\zeta_{i,j}^t)(y_i^t - y_j^t), \quad (72)$$

$$y_j^{t+1} = y_j^t - \lambda w_{i,j} f_a^{t-SNE}(\zeta_{i,j}^t)(y_i^t - y_j^t), \quad (73)$$

where f_a^{t-SNE} is the attraction shape of t -SNE and $\zeta_{i,j} = \|y_i - y_j\|_2$. The update equation for the repulsive parts are

$$y_i^{t+1} = y_i^t + \lambda \frac{w_{i,j}}{Z} \sum_k f_r^{t-SNE}(\zeta_{i,k})(y_i^t - y_k^t), \quad (74)$$

$$y_j^{t+1} = y_j^t - \lambda \frac{w_{i,j}}{Z} \sum_l f_r^{t-SNE}(\zeta_{l,j})(y_l^t - y_j^t), \forall j, j \neq i, \quad (75)$$

where f_r^{t-SNE} is the repulsion shape of t -SNE and $Z = \sum_{k \neq l} q_{k,l}$. The functional forms of these shapes are

$$f_a^{t-SNE}(\zeta) = -\frac{2}{1 + \zeta^2}, \text{ and} \quad (76)$$

$$f_r^{t-SNE}(\zeta) = \frac{2}{(1 + \zeta^2)^2}. \quad (77)$$

From the attractive update Eqs. (72)-(73), f_a^{t-SNE} follows Proposition 4.1 (with $0 < \lambda w_{i,j} f_a^{t-SNE} < -1$) and yields the minimum distance for contraction, ζ_{-1} . The repulsive update is coupled across all pairs and thus bears no simple relation to the repulsion shape. Building on our earlier analysis of TriMap, however, we can derive the following general condition:

Proposition G.3. *The update Eqs. (74)-(75) provide an expansion if*

$$h_r^{t-SNE}(\zeta_{i,j}, v, \theta, \lambda, w_{i,j}) > 1, \quad (78)$$

where

$$h_r^{t-SNE}(\zeta_{i,j}, v, \theta, \lambda, w_{i,j}) = (1 + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}))^2 + \frac{\|v\|_2^2}{\zeta_{i,j}^2} + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}) \frac{\|v\|_2}{\zeta_{i,j}} \cos \theta,$$

$$v = \lambda \frac{w_{i,j}}{Z} \left(\sum_{k,k \neq j} f_r^{t-SNE}(\zeta_{i,k})(y_i - y_k) + \sum_{l,l \neq i} f_r^{t-SNE}(\zeta_{l,j})(y_l - y_j) \right),$$

and θ is the angle between $(y_i - y_j)$ and v .

Proof. We require,

$$\|y_i^{t+1} - y_j^{t+1}\|_2^2 > \|y_i^t - y_j^t\|_2^2. \quad (79)$$

From Eqs. (74) and (75), $y_i - y_j = (1 + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}))(y_i - y_j) + v$. Putting this value in Eq. (79), we obtain the desired inequality. $\square$

Since t -SNE's condition for repulsion (Eq. 78) mirrors that of TriMap, the repulsion behavior is the same. Consequently, t -SNE's repulsive forces can produce both attraction and repulsion, just as in TriMap.

Because t -SNE's forces are scaled (by $w_{i,j}$ for attraction and $\frac{w_{i,j}}{Z}$ for repulsion), the attractive and repulsive forces are typically weaker than those of UMAP. As a result, the algorithm generally requires large learning rates (e.g., $\approx 10^3$ in the Open- t -SNE package (Poličar et al. 2024)). Moreover, most t -SNE implementations require an 'early exaggeration' step in which the attractive forces are multiplied by a constant for the first few iterations. This draws together points that should be close but are currently placed far apart (inducing far-sightedness). Conversely, if some points are very close ($\zeta_{i,j} \approx \zeta_{-1}$) but require separation, the early exaggeration trick accomplishes that as well. Thus, 'early exaggeration' plays a vital role in finding a consistent embedding in t -SNE and is an indispensable feature of the algorithm, especially under random initialization — an observation that our shape-based account explains in the same terms as the far-sightedness modifications used elsewhere in this paper. (Applied throughout the optimization, this trick makes t -SNE embeddings resemble UMAP embeddings (Böhm et al. 2022).)

G.5 Stochastic Neighbor Embedding (SNE)

SNE (Hinton and Roweis, 2002) is one of the oldest algorithms in this broad class. It follows the same formulation as t -SNE (Eq. 71) but with $q_{i,j} = \exp(-\|y_i - y_j\|_2^2)$. This yields the attraction shape

$$f_a^{SNE}(\zeta) = -2, \quad (80)$$

that follows Proposition 4.1 (with $0 < \lambda w_{i,j} f_a^{SNE} < -1$) and the repulsion shape

$$f_r^{SNE}(\zeta) = 2 \exp(-\zeta^2), \quad (81)$$

which follows Proposition G.3 (replacing the t -SNE symbols with their SNE counterparts). The attraction shape is ill-posed and thus relies primarily on the learning rate (λ) and the weights ($w_{i,j}$) for contraction, causing clusters to overlap even for a small number of samples (the so-called crowding problem (Van der Maaten and Hinton, 2008)). t -SNE and subsequent algorithms address this by moving from the Gaussian kernel to a heavy-tailed one, i.e., Eq. (43) — a transition our framework renders as a change in the underlying shape.

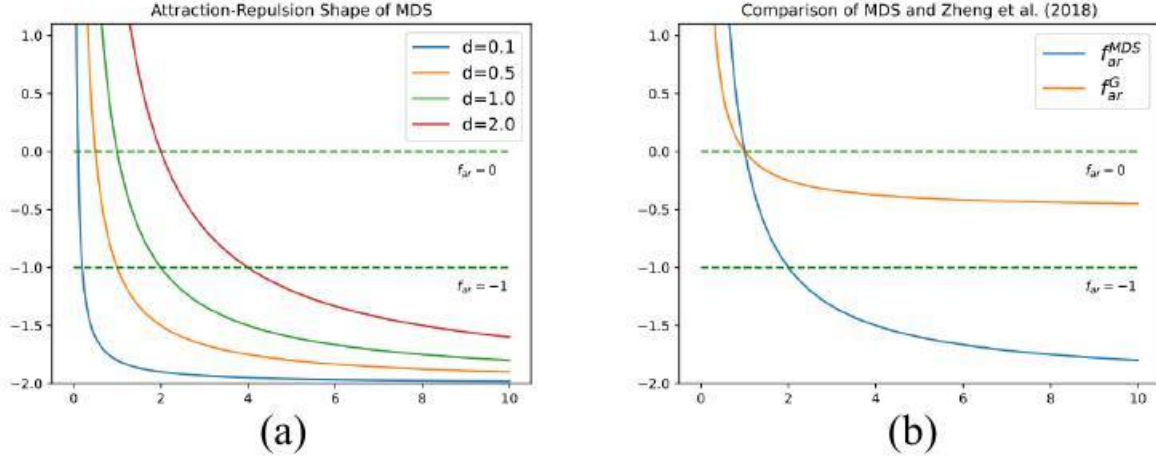


Figure 17: (a) Attraction-repulsion characteristic of MDS via f_{ar} . (b) Comparison of the attraction-repulsion characteristics of MDS and the method in Zheng et al. (2018) for $d = 1.0$.

G.6 Multidimensional Scaling (MDS)

Multidimensional scaling (MDS) (Borg and Groenen, 2007) typically avoids gradient methods, which often fail to converge to good mappings, and instead employs stress majorization. Nevertheless, a few works do consider gradient methods (Kruskal, 1964; Zheng et al., 2018), and we give a brief treatment below to show that even this comparatively distant method fits our framework. In particular, Zheng et al. (2018) formulates a successful gradient descent-based MDS algorithm for graph drawing. We begin with the cost function (the weight $w_{i,j}$ can be ignored without loss of generality with respect to their approach, and for the present discussion it can be absorbed into the learning rate):

$$\mathcal{L}^{MDS} = \sum_{i,j,i < j} (\|y_i - y_j\| - d_{ij})^2. \quad (82)$$

The loss has a single term per pair and lacks explicit attractive and repulsive terms; the term itself supplies both attraction and repulsion. Fortunately, the loss is separable into individual (i, j) components, so we can analyze a single pair. The derivative is given by

$$\frac{\partial}{\partial y_i} (\|y_i - y_j\| - d_{ij})^2 = 2 \frac{\|y_i - y_j\| - d_{ij}}{\|y_i - y_j\|} (y_i - y_j). \quad (83)$$

Thus, the attraction-repulsion shape of MDS becomes

$$f_{ar}^{MDS}(\zeta) = -2 \frac{\zeta - d}{\zeta}, \quad (84)$$

where $\zeta = \|y_i - y_j\|_2$. Here, the shape is attractive when $\zeta > d$. In this range, the shape is bounded within $[-2, 0]$ (Fig. 17(a)). This is not well suited to convergence, however, since only values within $[-1, 0]$ contract while the rest expand. Worse, values below -2 cause the points to flip and expand. To work around this, Zheng et al. (2018) uses an ad hoc formulation inspired by force-directed graphs, given by

$$f_{ar}^G(\zeta) = -\frac{1}{2} \frac{\zeta - d}{\zeta}, \quad (85)$$

which works for an effective learning rate $\lambda \leq 1$. Since f_{ar}^G is bounded within $[-0.5, 0]$, the choice $\lambda \leq 1$ suffices. For $\zeta < d$, on the other hand, $f_{ar}^{MDS}(> 0)$ behaves repulsively. Overall, this attraction-repulsion interaction works best when the initialization is already close to the desired output. Continued optimization of a pair makes it oscillate around the distance ($\zeta = d$) where $f_{ar}^G = 0$, giving a notion analogous to UMAP's ζ_{-1} . No choice of learning rate drives this distance to zero (indeed, doing so would defeat the objective of MDS). As a result, the clusters are often fuzzy compared to methods such as UMAP and PaCMAP (for relevant illustrations, see Lambert et al. (2022), de Bodt et al. (2025), and Kury et al. (2025)), a difference that the shape framework makes precise even across this broad range of methods. Note that Lambert et al. (2022) converts the MDS loss into a quartet stress (a loss involving four samples) and a relative distance (distance divided by the sum of six distances in the quartet), enabling global regularization and reduced computation relative to the majorization approach.

H Construction of the High-Dimensional Graph

Stochastic Neighbor Embedding (SNE) [Hinton and Roweis, 2002] underpins the broad family of modern dimensionality reduction algorithms. It constructs a high-dimensional graph of the dataset $X = \{x_i \in R^n | i = 1, \dots, N\}$ via the following system of equations:

$$w_{ij} = \frac{p_{j|i} + p_{i|j}}{2N}, \quad (86)$$

$$w_{j|i} = \frac{\exp(-\|x_i - x_j\|_2^2 / 2\sigma_i^2)}{\sum_{t \neq v} \exp(-\|x_t - x_v\|_2^2 / 2\sigma_t^2)}, \quad (87)$$

where σ_i^2 is chosen to match a user-defined value *perplexity*, P , defined as

$$P = 2^{H_i} \quad (88)$$

$$H_i = \sum_{j \neq i} w_{j|i} \log_2 w_{j|i}. \quad (89)$$

UMAP, by contrast, constructs its high-dimensional graph through the following system of equations, relying on the k -nearest neighbor (k -NN) algorithm:

$$p_{i|j} = \begin{cases} \exp\left(-\frac{d(x_i, x_j) - \rho_i}{\sigma_i}\right) & \text{if } x_j \in \text{KNN}(x_i, k) \\ 0 & \text{otherwise} \end{cases}, \quad (90)$$

$$\rho_i = \min_{x_j \in \text{KNN}(x_i, k)} d(x_i, x_j), \quad (91)$$

where $\text{KNN}(x_i, k)$ is the set of k -nearest neighbors of the point x_i and σ_i is a scaling parameter such that $\sum_j p_{i|j} = \log_2(k)$. The graph is then symmetrized by a t-conorm:

$$p_{i,j} = p_{i|j} + p_{j|i} - p_{i|j}p_{j|i}. \quad (92)$$

PaCMAP uses just the k -NN graph for the affinities (all equal to 1) with a self-tuning distance measure [Zelnik-Manor and Perona, 2004],

$$d_{i,j}^2 = \frac{\|x_i - x_j\|^2}{\sigma_i \sigma_j}, \quad (93)$$

where σ_i is the average distance between x_i and its fourth-through-sixth Euclidean nearest neighbors. This serves the same purpose as the corresponding σ_i parameters in Eqs. (87) and (90), even though all three are defined differently.

Regardless of these particular choices, the optimization — and, by our analysis, the attraction and repulsion shapes — remains the primary driver of the low-dimensional embedding across all these methods.

I Detailed Results for Varying λ_a and λ_b

In this section, we provide detailed results for the experiments in Fig. 11(h,i). We obtained these results by changing the attraction and repulsion shapes of UMAP to those of the other methods. Figure 18 reproduces the results reported in the main text. Figures 19-22 show the individual embeddings for each choice of λ_a and λ_r for every method, along with their initialization, a reference embedding when the learning rate λ is annealed, and the cases where either λ_a or λ_r is set to 0. For PaCMAP, which uses the concept of mid-near pairs, we additionally include a reference with mid-near pairs. To demonstrate that these detailed trends generalize beyond a single dataset, Figures 23-27 and Figures 28-32 provide the corresponding results on the FMNIST and single-cell transcriptomes datasets, respectively.

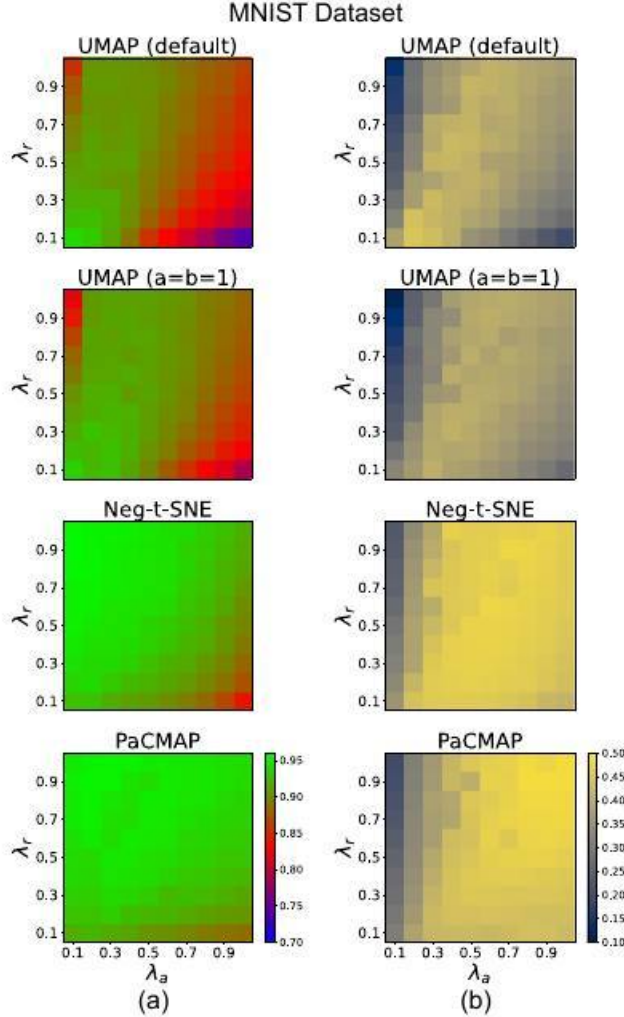


Figure 18: (a) Trustworthiness and (b) silhouette score across the different methods for the MNIST dataset.

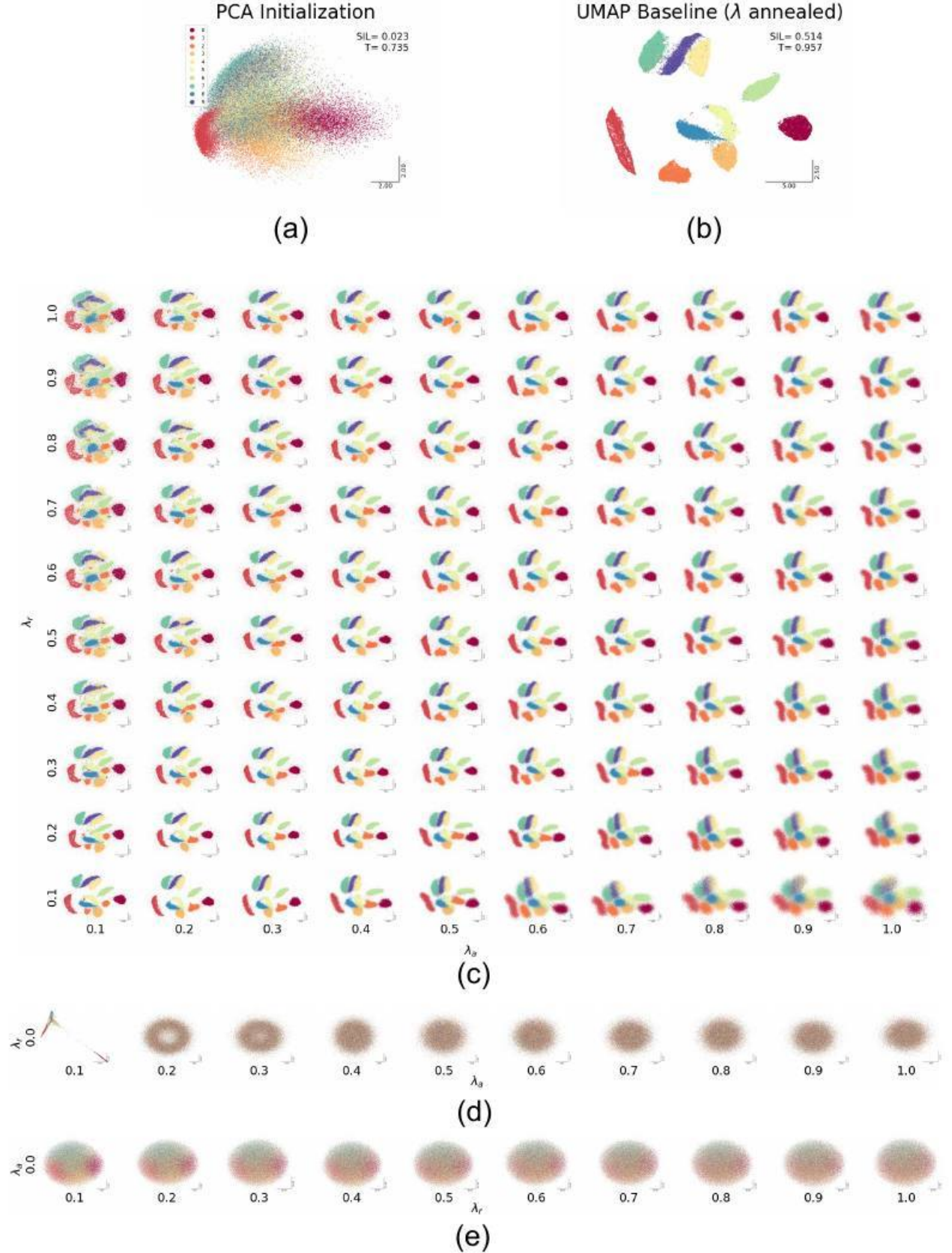


Figure 19: Varying λ_a and λ_r for the MNIST dataset using UMAP's attraction and repulsion shapes; the same qualitative pattern recurs for the other methods and datasets below. (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) Embeddings as λ_a and λ_r vary (without any annealing). (d) With λ_r set to 0 (no repulsion), attraction alone cannot produce any cluster. (e) Likewise, with λ_a set to 0 (no attraction), repulsion alone cannot produce any clusters.

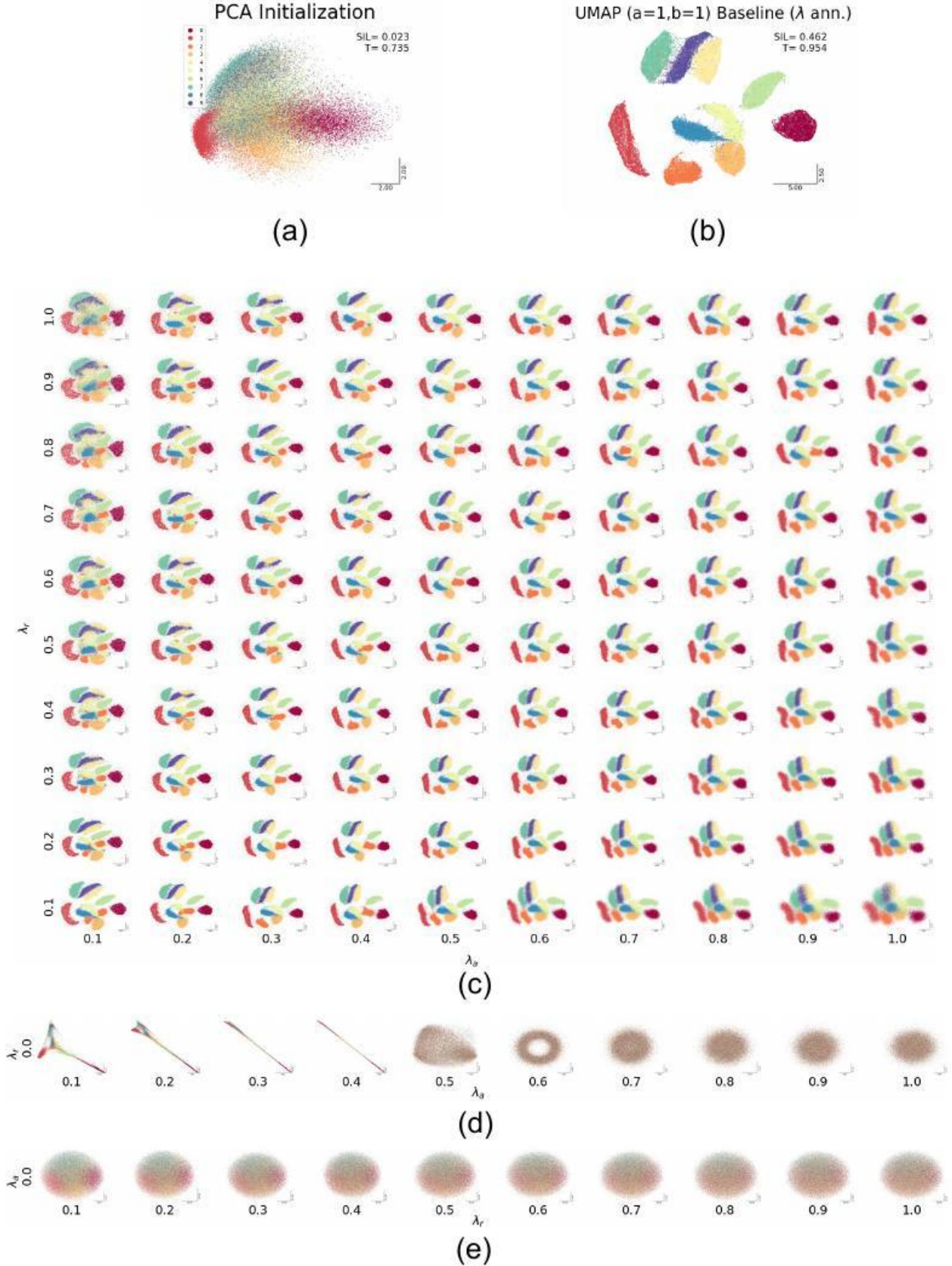


Figure 20: Varying λ_a and λ_r for the MNIST dataset using UMAP's attraction and repulsion shapes (with $a = 1$ and $b = 1$). (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) Embeddings as λ_a and λ_r vary (without any annealing). (d) With λ_r set to 0 (no repulsion), attraction alone cannot produce any cluster. (e) Likewise, with λ_a set to 0 (no attraction), repulsion alone cannot produce any clusters.

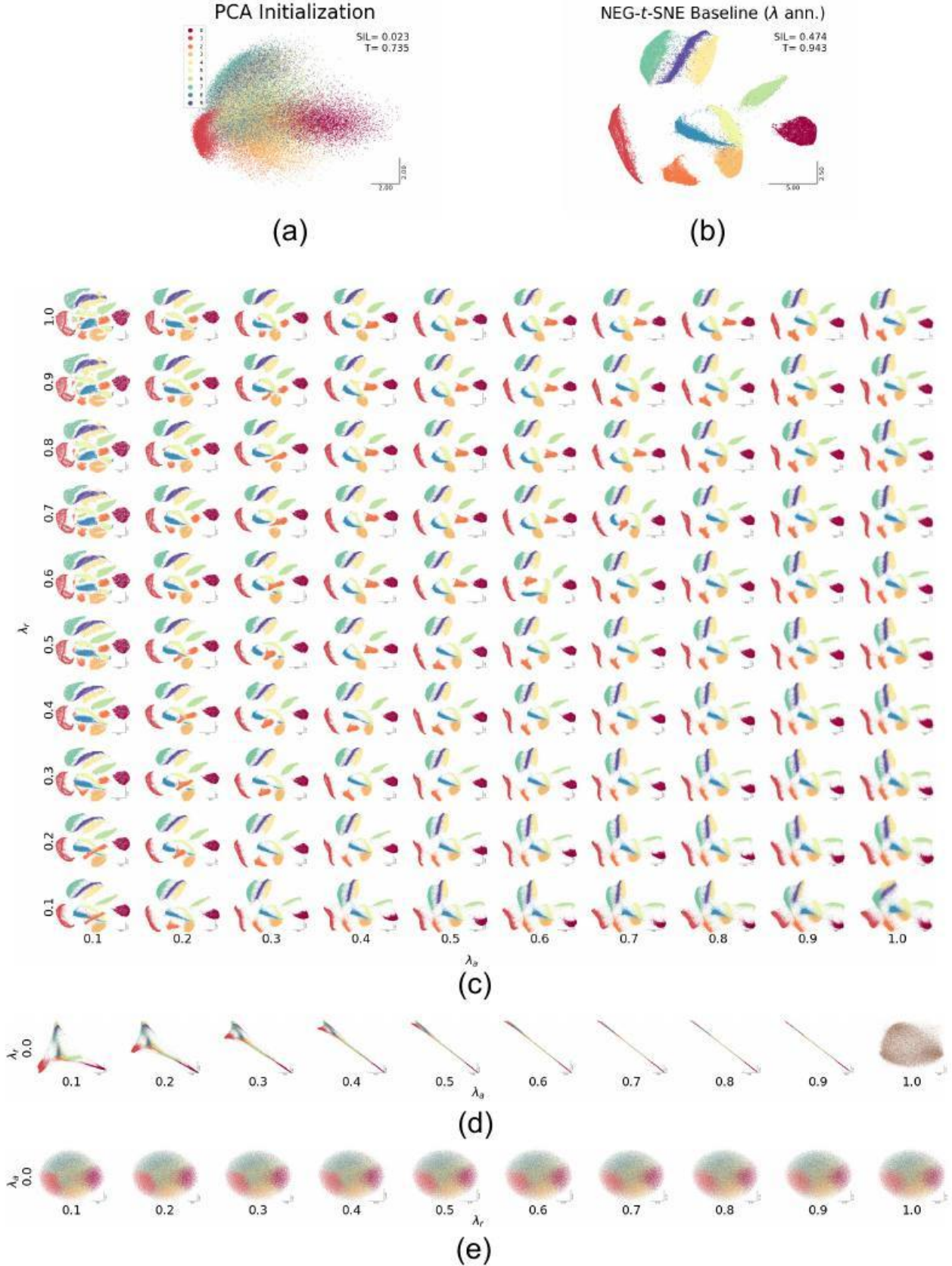


Figure 21: Varying λ_a and λ_r for the MNIST dataset using NEG-t-SNE's attraction and repulsion shapes. (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) Embeddings as λ_a and λ_r vary (without any annealing). (d) With λ_r set to 0 (no repulsion), attraction alone cannot produce any cluster. (e) Likewise, with λ_a set to 0 (no attraction), repulsion alone cannot produce any clusters.

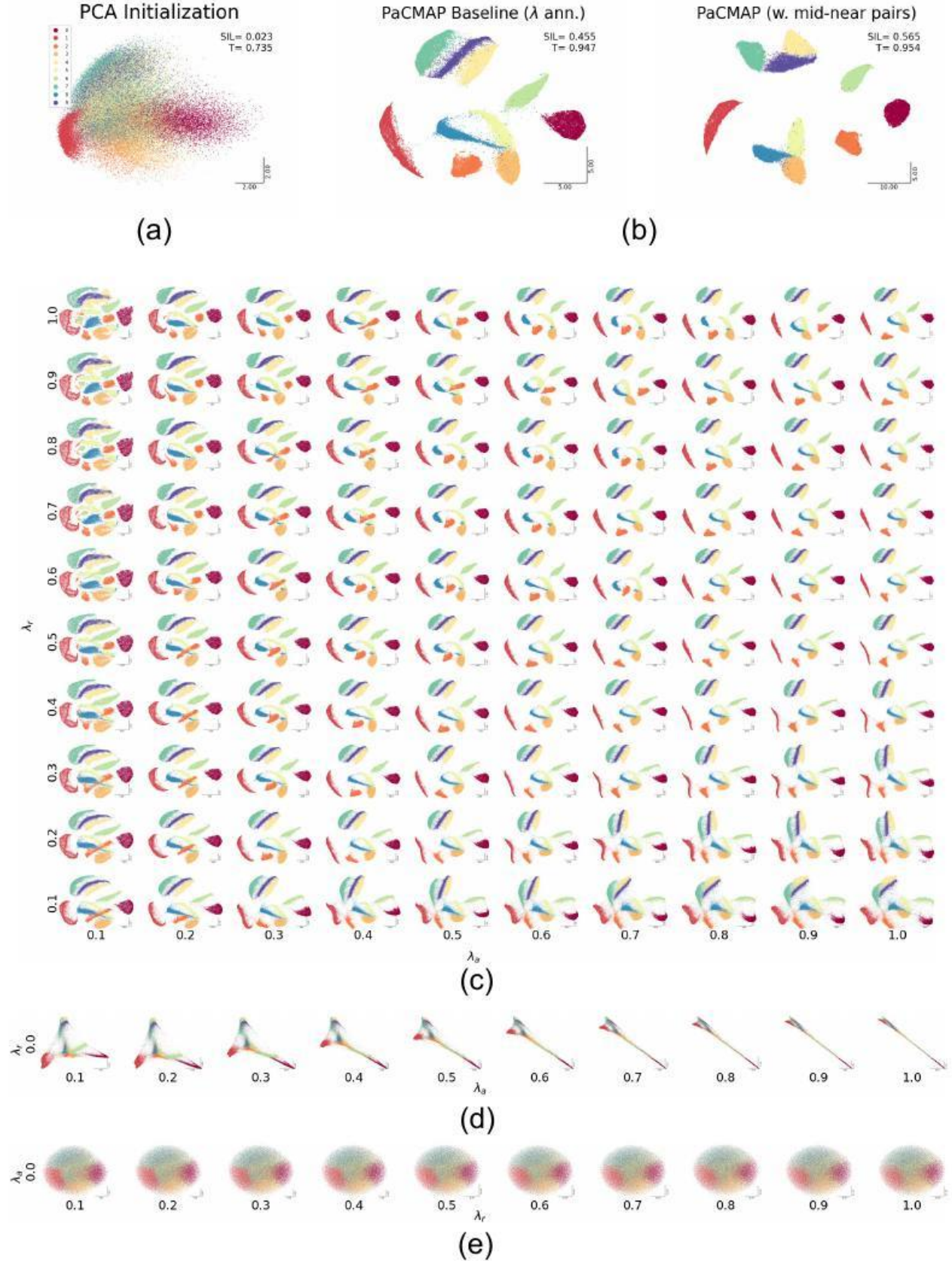


Figure 22: Varying λ_a and λ_r for the MNIST dataset using PaCMAP's attraction and repulsion shapes. (a) Initialization for the embeddings. (b) Left: baseline when λ is annealed from 1 (mid-near points excluded so as to isolate the attraction-repulsion interaction); right: with mid-near points included. (c) Embeddings as λ_a and λ_r vary (without any annealing). (d) With λ_r set to 0 (no repulsion), attraction alone cannot produce any cluster. (e) Likewise, with λ_a set to 0 (no attraction), repulsion alone cannot produce any clusters.

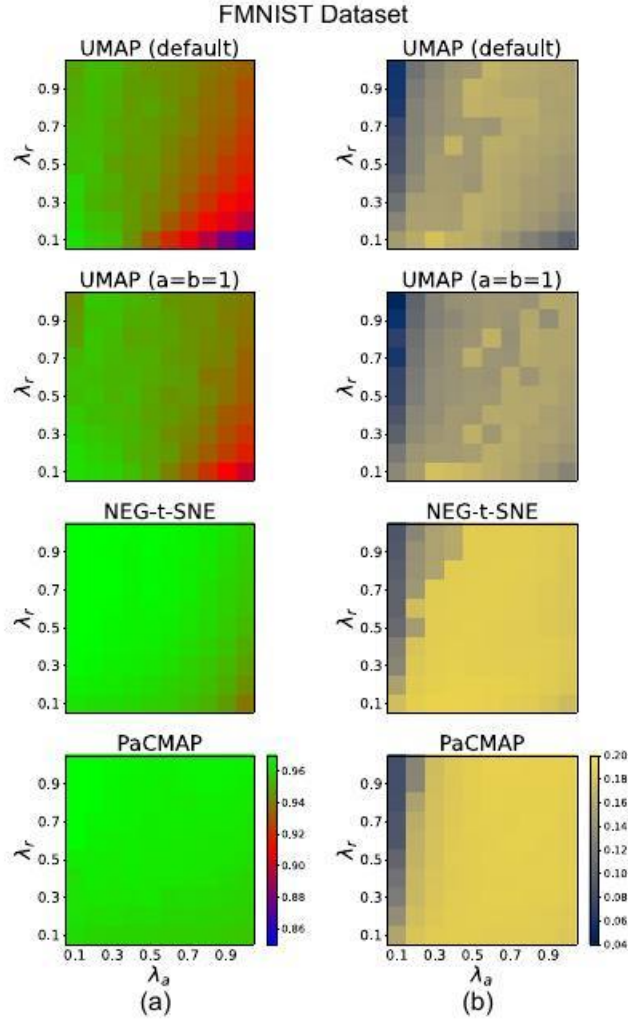


Figure 23: (a) Trustworthiness and (b) silhouette score across the different methods for the FMNIST dataset.

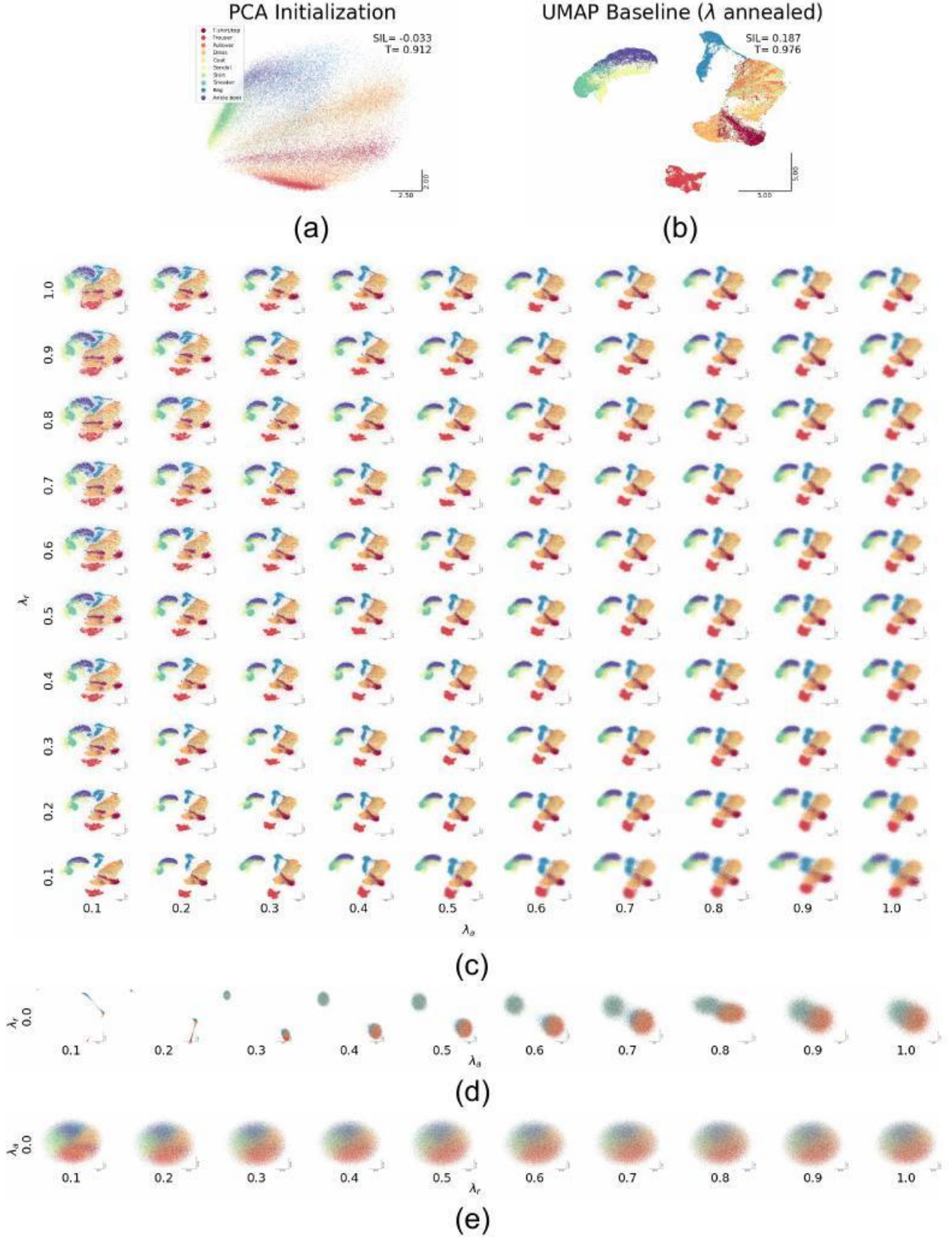


Figure 24: Varying λ_a and λ_r for the FMNIST dataset using UMAP’s attraction and repulsion shapes, confirming that the MNIST trends carry over to another image domain. (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) Embeddings as λ_a and λ_r vary (without any annealing). (d) With λ_r set to 0 (no repulsion), attraction alone cannot produce any cluster. (e) Likewise, with λ_a set to 0 (no attraction), repulsion alone cannot produce any clusters.

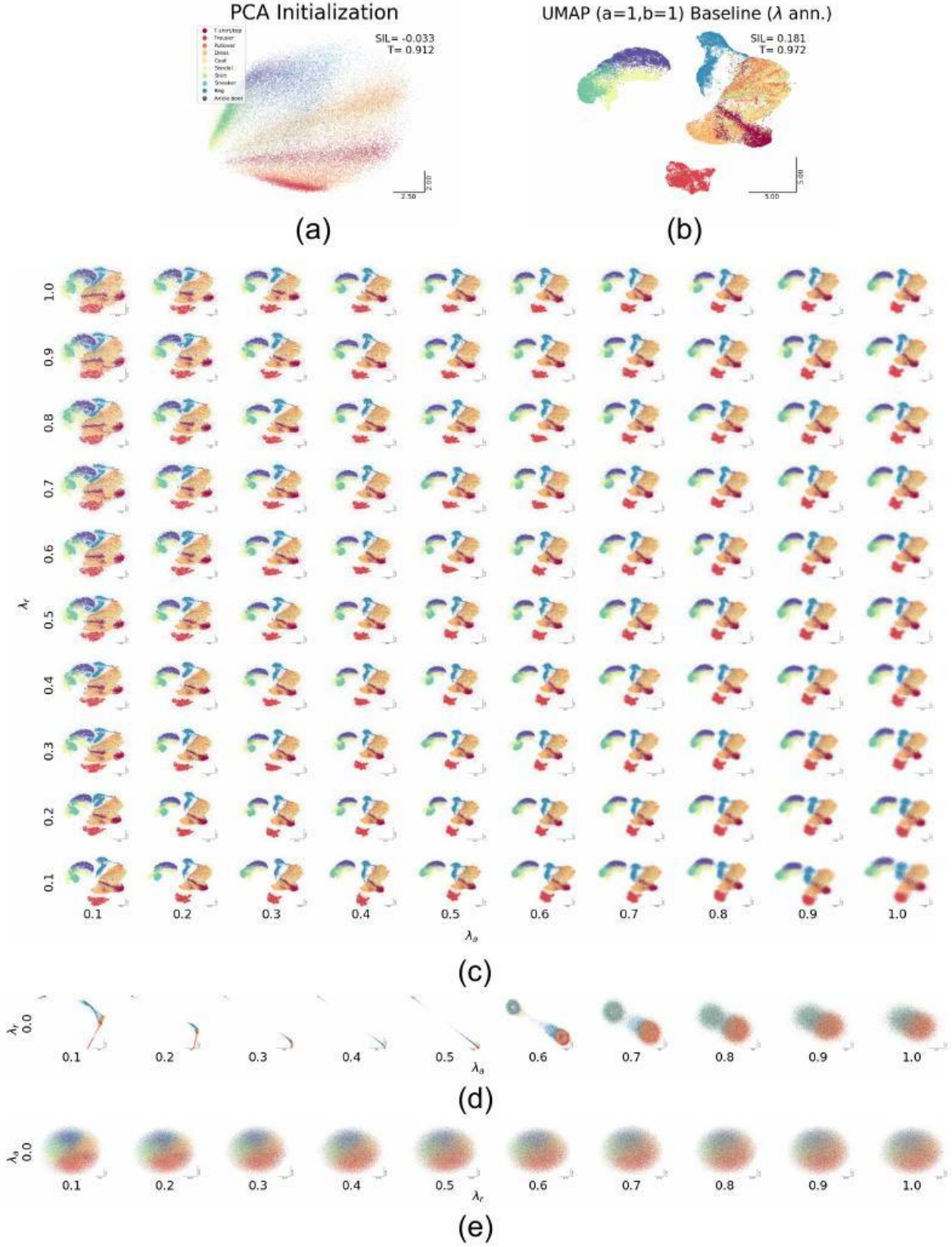


Figure 25: Varying λ_a and λ_r for the FMNIST dataset using UMAP's attraction and repulsion shapes (with $a = 1$ and $b = 1$). (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) Embeddings as λ_a and λ_r vary (without any annealing). (d) With λ_r set to 0 (no repulsion), attraction alone cannot produce any cluster. (e) Likewise, with λ_a set to 0 (no attraction), repulsion alone cannot produce any clusters.

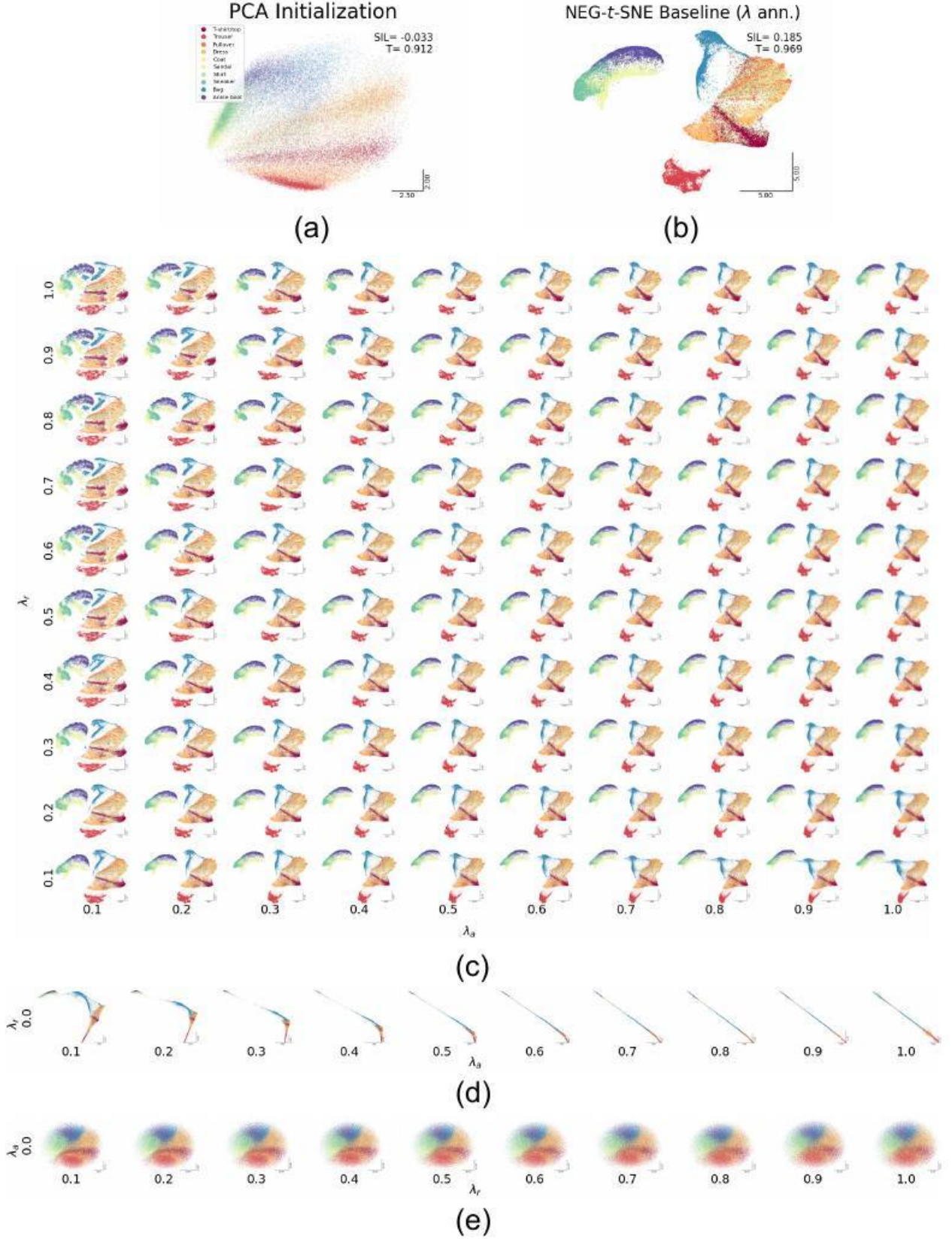


Figure 26: Varying λ_a and λ_r for the FMNIST dataset using NEG-t-SNE's attraction and repulsion shapes. (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) Embeddings as λ_a and λ_r vary (without any annealing). (d) With λ_r set to 0 (no repulsion), attraction alone cannot produce any cluster. (e) Likewise, with λ_a set to 0 (no attraction), repulsion alone cannot produce any clusters.

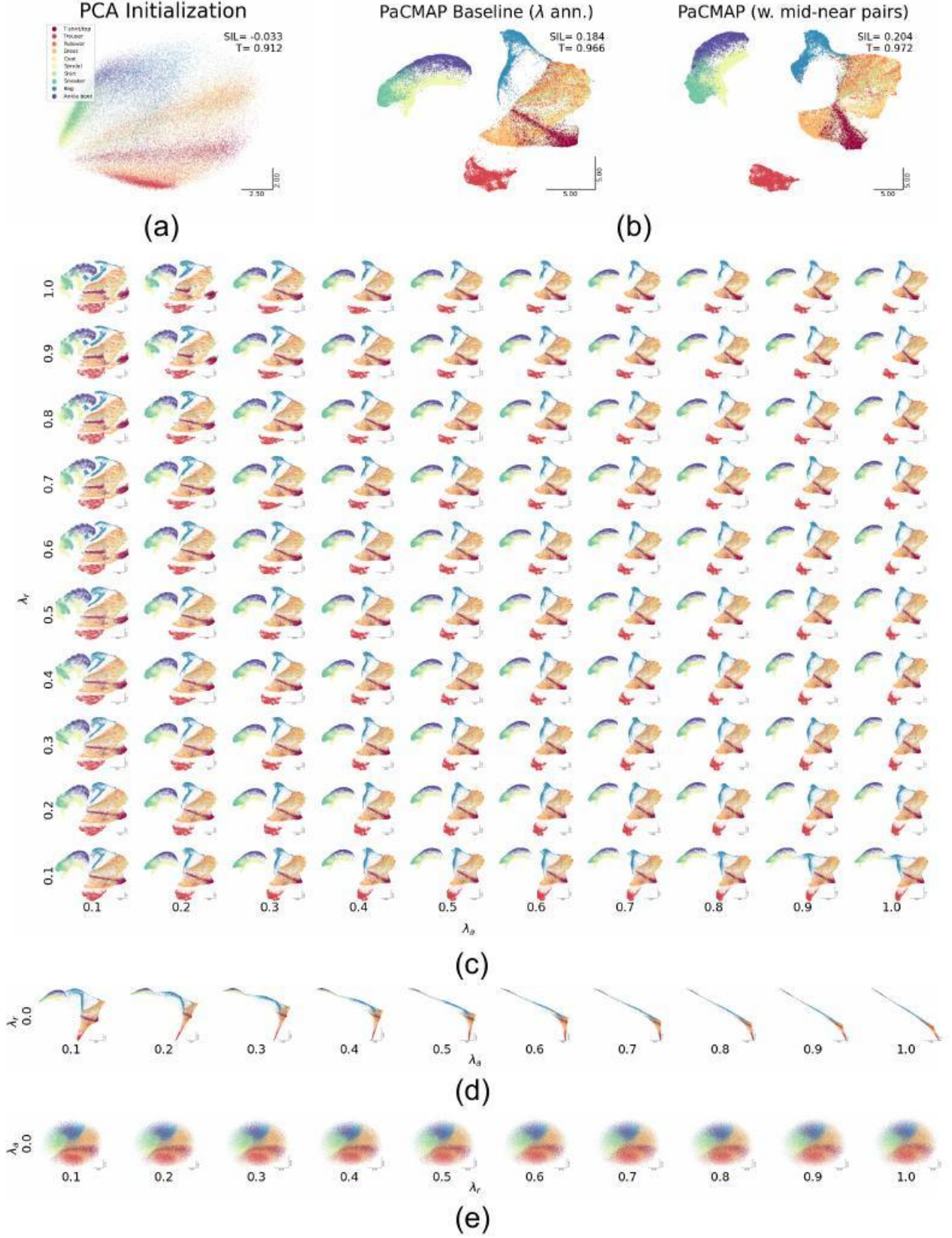


Figure 27: Varying λ_a and λ_r for the FMNIST dataset using PaCMAP's attraction and repulsion shapes. (a) Initialization for the embeddings. (b) Left: baseline when λ is annealed from 1 (mid-near points excluded so as to isolate the attraction-repulsion interaction); right: with mid-near points included. (c) Embeddings as λ_a and λ_r vary (without any annealing). (d) With λ_r set to 0 (no repulsion), attraction alone cannot produce any cluster. (e) Likewise, with λ_a set to 0 (no attraction), repulsion alone cannot produce any clusters. 40

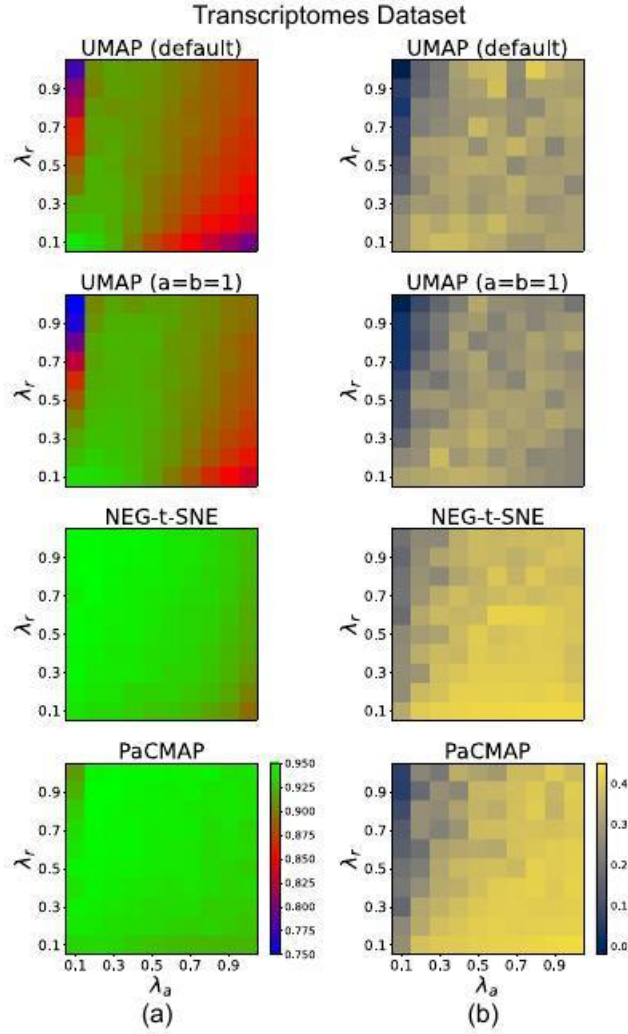


Figure 28: (a) Trustworthiness and (b) silhouette score across the different methods for the single-cell transcriptomes dataset.

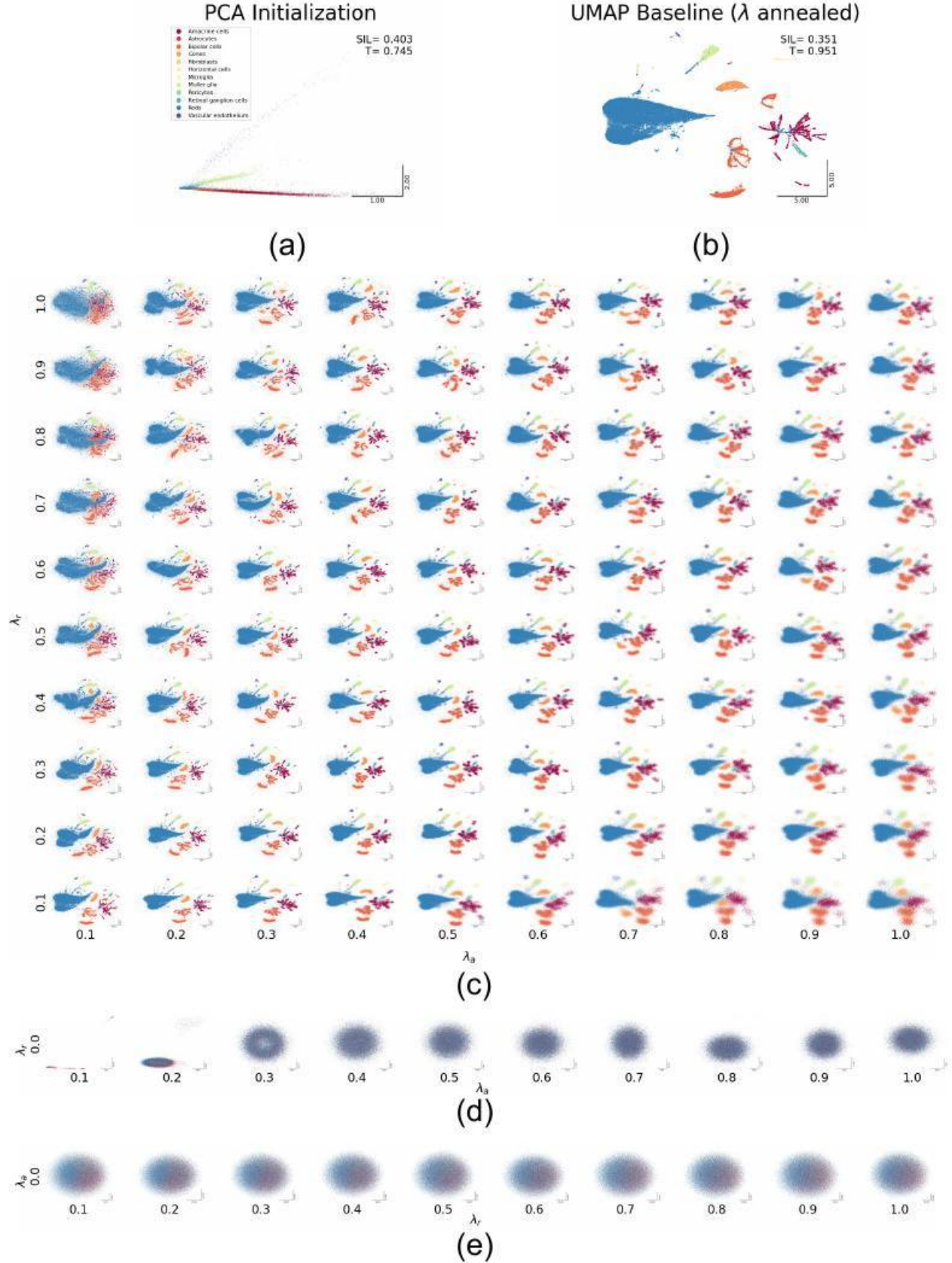


Figure 29: Varying λ_a and λ_r for the single-cell transcriptomes dataset using UMAP's attraction and repulsion shapes, showing that the trends extend from image data to biological data. (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) Embeddings as λ_a and λ_r vary (without any annealing). (d) With λ_r set to 0 (no repulsion), attraction alone cannot produce any cluster. (e) Likewise, with λ_a set to 0 (no attraction), repulsion alone cannot produce any clusters.

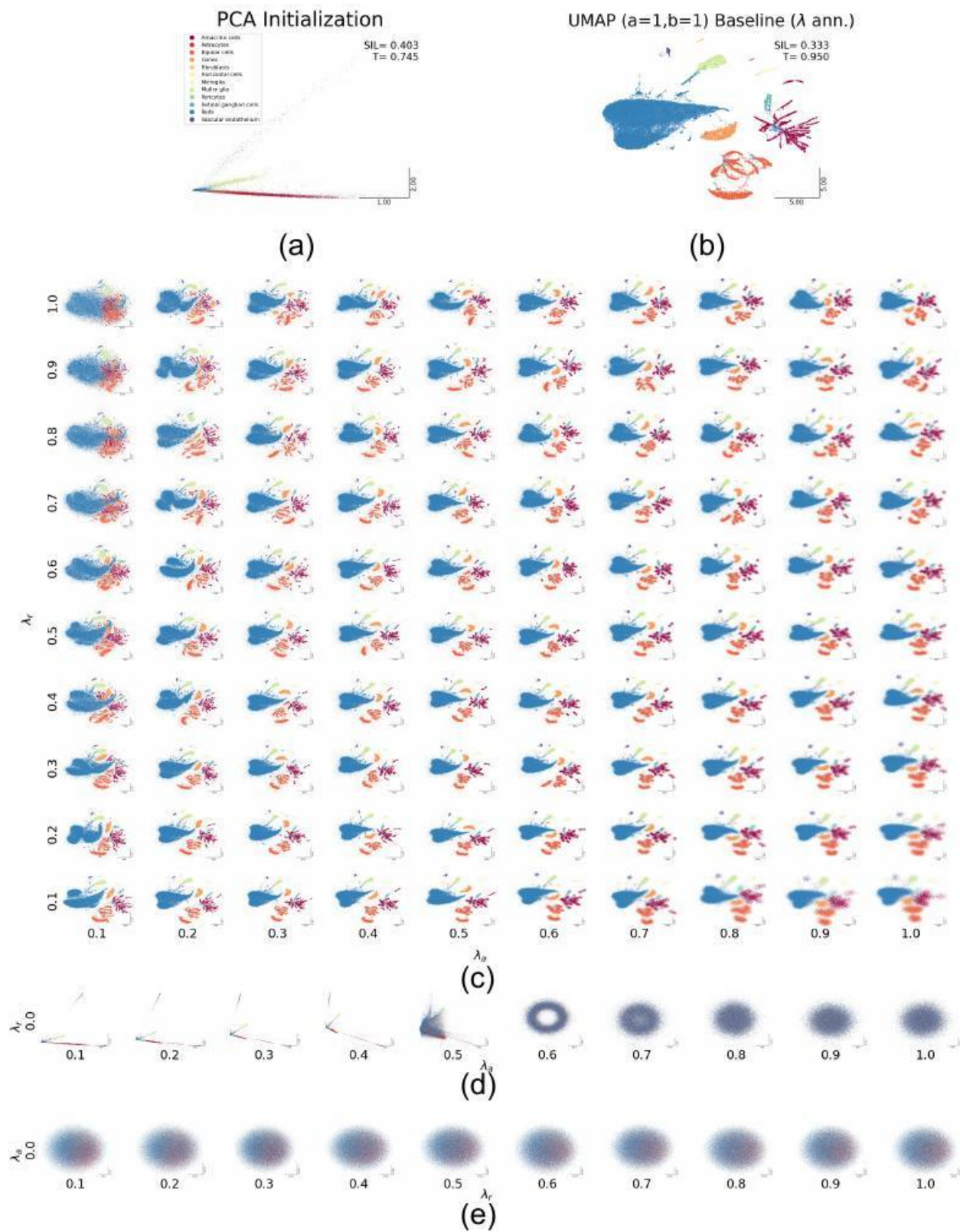


Figure 30: Varying λ_a and λ_r for the single-cell transcriptomes dataset using UMAP’s attraction and repulsion shapes (with $a = 1$ and $b = 1$). (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) Embeddings as λ_a and λ_r vary (without any annealing). (d) With λ_r set to 0 (no repulsion), attraction alone cannot produce any cluster. (e) Likewise, with λ_a set to 0 (no attraction), repulsion alone cannot produce any clusters.

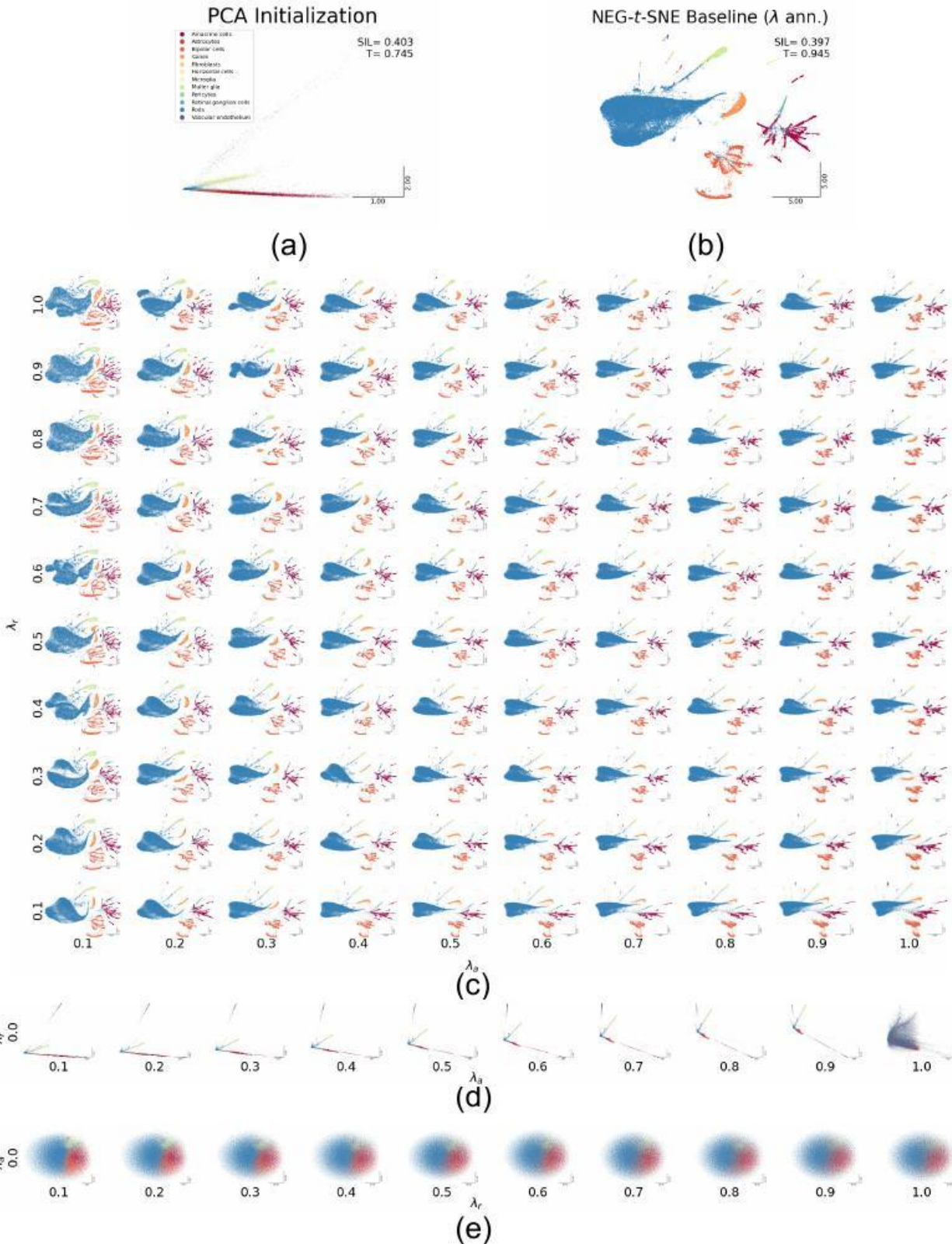


Figure 31: Varying λ_a and λ_r for the single-cell transcriptomes dataset using NEG-*t*-SNE’s attraction and repulsion shapes. (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) Embeddings as λ_a and λ_r vary (without any annealing). (d) With λ_r set to 0 (no repulsion), attraction alone cannot produce distinct clusters. (e) Likewise, with λ_a set to 0 (no attraction), repulsion alone cannot produce any clusters.

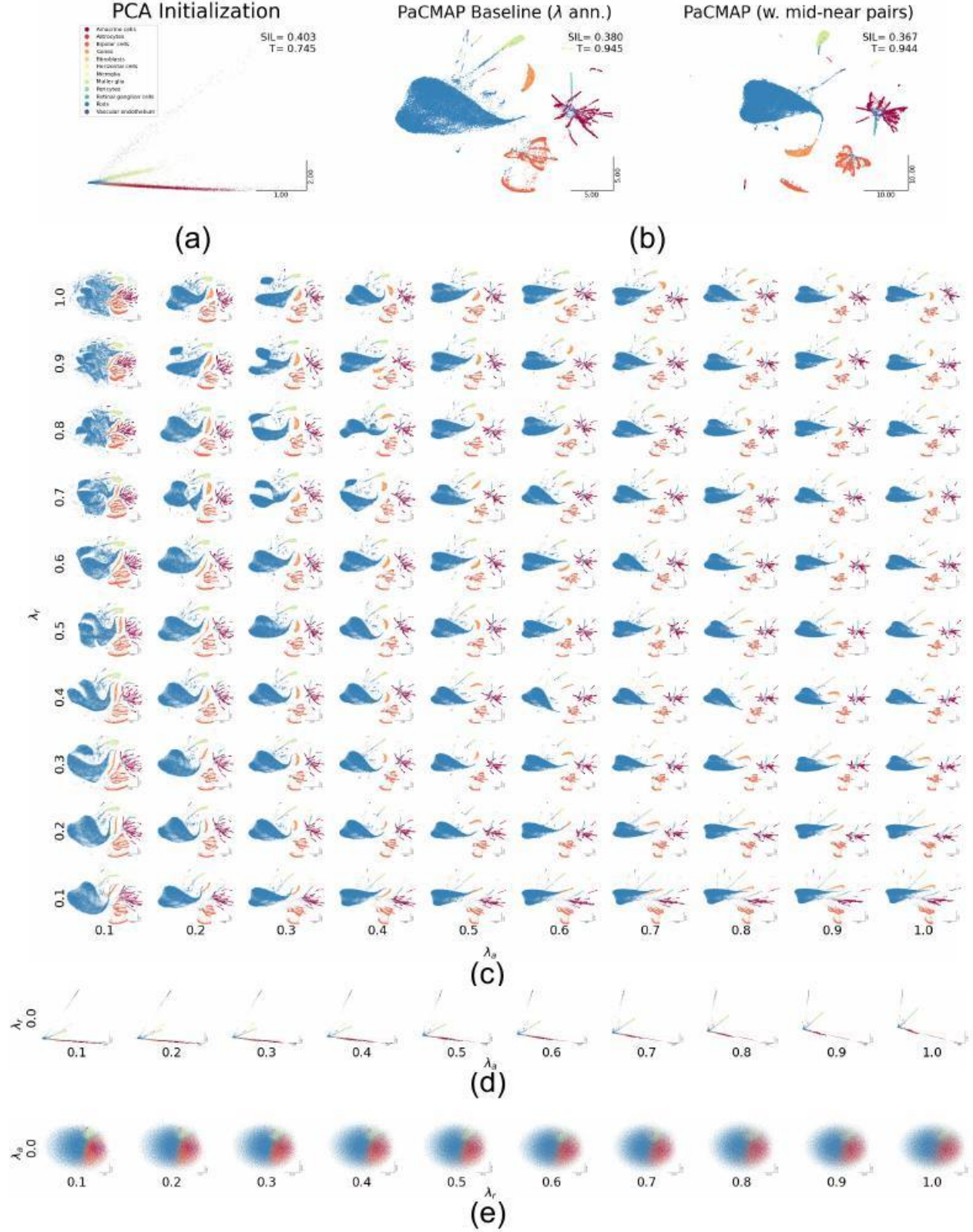


Figure 32: Varying λ_a and λ_r for the single-cell transcriptomes dataset using PaCMAP's attraction and repulsion shapes. (a) Initialization for the embeddings. (b) Left: baseline when λ is annealed from 1 (mid-near points excluded so as to isolate the attraction-repulsion interaction); right: with mid-near points included. (c) Embeddings as λ_a and λ_r vary (without any annealing). (d) With λ_r set to 0 (no repulsion), attraction alone cannot produce any cluster. (e) Likewise, with λ_a set to 0 (no attraction), repulsion alone cannot produce any clusters.

J Mix and Match Approach

Throughout our analysis, we treated the attraction and repulsion shapes as independent of each other. This independence implies a broadly useful degree of freedom: attraction and repulsion shapes derived from different loss functions and principles can be freely combined. Figure 33 shows several such examples. Here, we restricted the attraction shapes to $[-1, 0]$ (to satisfy Proposition 4.1) so that they become less sensitive to learning rates, and we used finite repulsion shapes, thereby mitigating the effect of large repulsive forces at small distances. We held the learning rate constant ($\lambda = 1$) and used the same (UMAP) optimization for all of them, so that any differences arise from the shapes themselves rather than from learning-rate scheduling or the optimization scheme. Overall, UMAP’s attraction shape (via $f_a = 0.5f_a^U$) produces better clusters (higher silhouette score), while PaCMAP’s attraction shape yields better local structure (higher trustworthiness). The latter follows primarily from PaCMAP’s attraction shape lying within $[-0.5, 0]$, which prevents any flips during the updates. By confining UMAP’s and NEG- t -SNE’s attraction shapes to $[-0.5, 0]$ (via $f_a = 0.25f_a^U$ and $f_a = 0.5f_a^N$, respectively), we can improve local structure at the expense of clustering performance (Fig. 34). These examples demonstrate that the shapes act as interchangeable, method-agnostic building blocks.

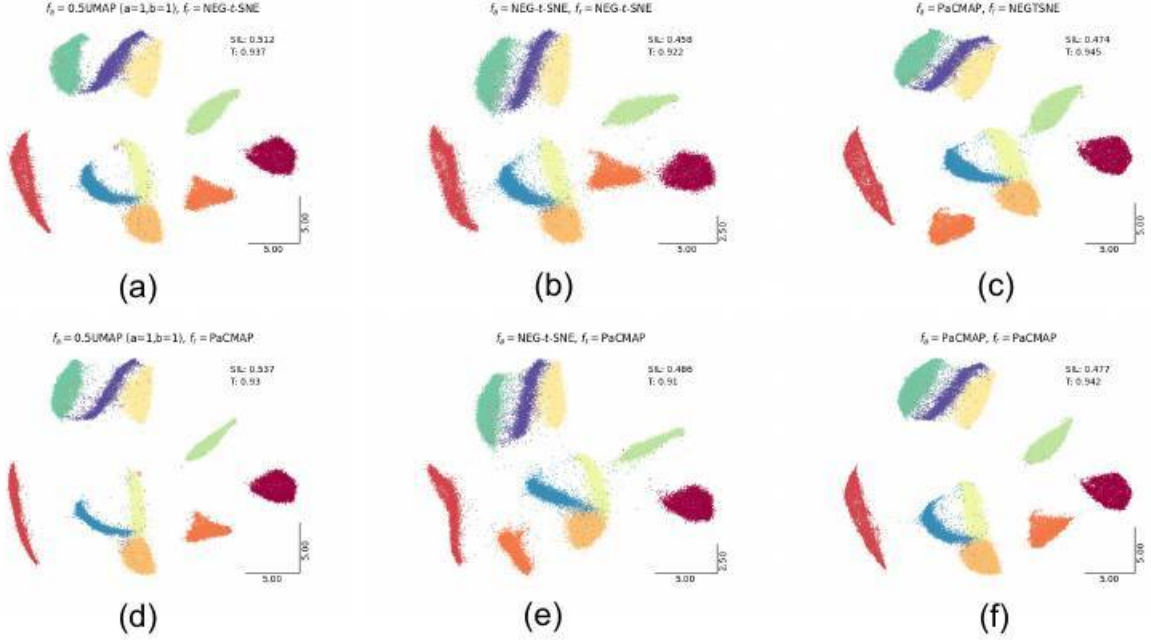


Figure 33: Embedding of MNIST obtained by mixing and matching attraction and repulsion shapes drawn from different methods, all at a constant learning rate ($\lambda = 1$). We used attraction shapes satisfying Proposition 4.1 and finite repulsion shapes. Top row: UMAP, NEG- t -SNE, and PaCMAP attraction shapes paired with NEG- t -SNE’s repulsion shape. Bottom row: UMAP, NEG- t -SNE, and PaCMAP attraction shapes paired with PaCMAP’s repulsion shape.

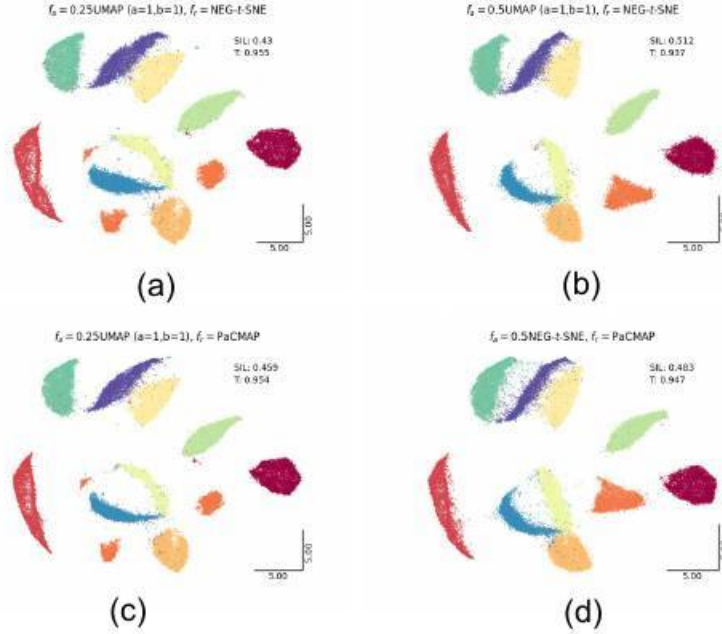


Figure 34: Embedding of MNIST obtained by mixing and matching attraction and repulsion shapes across methods, at a constant learning rate ($\lambda = 1$). Here we confined the attraction shape to $[-0.5, 0]$ to prevent any flips during the attractive update. Top row: UMAP and NEG- t -SNE attraction shapes paired with NEG- t -SNE’s repulsion shape. Bottom row: UMAP and NEG- t -SNE attraction shapes paired with PaCMAP’s repulsion shape.

K Implementation Details

For our analysis, we implemented our own UMAP algorithm. We used `numba` (Lam et al. 2015) to compute an exact (rather than approximate) nearest neighbor graph with $k = 15$, and `scikit-learn`'s (Pedregosa et al. 2011) implementation of PCA for PCA initialization. This implementation produced and quantified the embeddings in Figs. 1, 2, 3, 4, 6, 8, 9, and 10. The same implementation, with the attraction and repulsion shapes swapped for those of the alternative methods, produced the results in Figs. 18, 34 (unless otherwise stated), which lets us compare methods on equal footing. The trustworthiness and silhouette scores were computed with the corresponding functions from the `scikit-learn` package.

The mappings shown in Figs. 2, 3, 9, 10, and 15 are rotated to a reference embedding (panel (a) in each respective figure). To do so, we performed Procrustes alignment of the embeddings by normalizing them (zero mean and unit norm) and then using SciPy's (Virtanen et al. 2020) `orthogonal_procrustes` method to extract the rotation and scaling parameters.

For the comparison with NEG-*t*-SNE in Fig. 11, we used the original implementation of the contrastive embedding framework for both the UMAP and NEG-*t*-SNE algorithms; the implementation details will be released upon acceptance.

The PaCMAP and LocalMAP embeddings in Figs. 15, 16, 22 (b) (right), 27 (b) (right), and 32 (b) (right) were obtained using the official PaCMAP package; the implementation details will be released upon acceptance.

The code for displaying the embeddings of the PubMed dataset in Fig. 7 will be released upon acceptance.